Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Meirinho Holsteins Inc

Physical address of dairy:

Number and Street	City	County	Zip Code
4890 Healy RD	Merced	Merced	95341

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1962

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X066-X200-X019-XXXX

B. OPERATORS

Meirinho, Larry

Operator name: Meirinho, Larry

Telephone no.: (209) 578-1403 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5418 Woodland	Modesto	CA	95385

This operator is responsible for paying permit fees.

Meirinho, Tony

Operator name: Meirinho, Tony

Telephone no.: (209) 652-9780 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 580010	Modesto	CA	95357

C. OWNERS

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Meirinho, Larry

Legal owner name: Meirinho, Larry
Telephone no.: (209) 578-1403
Landline | Cellular

Mailing Address Number and Street	City	State	Zip Code
5418 Woodland	Modesto	CA	95385

This owner is responsible for paying permit fees.

Meirinho, Tony

Legal owner name: Meirinho, Tony
Telephone no.: (209) 652-9780
Landline | Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 580010	Modesto	CA	95357

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	1,230	940	440	450
Number under roof	3,230	500	0	0	0	0
Maximum number	3,300	600	1,300	1,000	500	600
Average number	3,230	500	1,230	940	440	450
Avg live weight (lbs)	1,400	1,400	900	650		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 116,251.66 *tons per reporting period*

Total nitrogen from manure: 1,450,721.25 *lbs per reporting period*

After ammonia losses (30% loss applied): 1,015,504.88 *lbs per reporting period*

Total phosphorus from manure: 238,060.26 *lbs per reporting period*

Total potassium from manure: 629,185.35 *lbs per reporting period*

Total salt from manure: 1,635,820.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 173,880,000 *gallons*

Total nitrogen generated: 493,227.39 *lbs*

Total phosphorus generated: 88,614.90 *lbs*

Total potassium generated: 659,524.29 *lbs*

Total salt generated: 3,683,572.29 *lbs*

	129,780,000	*gallons applied*
+	44,100,000	*gallons exported*
-	0	*gallons imported*
=	173,880,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Supply Well	Ground water
IW A	Ground water
IW B	Ground water
IW C	Ground water
IW D	Ground water

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Source Description	Type
IW L	Ground water
MID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	Compost	1,785.00 *ton*	Dry-weight	54.2		28,400.00	4,000.00	7,500.00		18.80

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
01/01/2023	Process wastewater	8,300,000.00 *gal*	293.00	218.00	0.00	1.80	47.60	328.00		1,930
03/07/2023	Process wastewater	9,500,000.00 *gal*	430.00	292.00	0.00	2.20	74.90	483.00		2,850
06/07/2023	Process wastewater	7,650,000.00 *gal*	391.00	352.00	0.00	21.90	75.30	563.00		3,140
06/13/2023	Process wastewater	6,500,000.00 *gal*	391.00	352.00	0.00	21.90	75.30	563.00		3,140
07/23/2023	Process wastewater	6,750,000.00 *gal*	266.00	236.00	0.00	0.30	54.50	414.00		2,150
08/01/2023	Process wastewater	5,400,000.00 *gal*	266.00	236.00	0.00	0.30	54.50	414.00		2,150

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	46,435.70	6,540.24	12,262.95	307,391.28
Process wastewater	130,439.19	23,652.23	169,465.67	948,388.39
Total exports for all materials	176,874.90	30,192.47	181,728.62	1,255,779.67

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	70	70	2	process wastewater	X066-X170-X027-XXXX
10	23	23	2	both	X066-X200-X013-XXXX
11	32	32	2	process wastewater	X066-X200-X013-XXXX
12	30	30	2	process wastewater	X066-X170-X028-XXXX
13	75	75	2	both	X066-X170-X028-XXXX
2	74	74	2	both	X066-X170-X027-XXXX
3	16	16	2	both	X066-X170-X027-XXXX
4	50	50	2	both	X066-X200-X020-XXXX
5	120	118	3	both	X066-X200-X020-XXXX
6	94	94	3	both	X066-X200-X021-XXXX
7	70	70	2	both	X066-X200-X027-XXXX
8	104	104	2	process wastewater	X066-X200-X027-XXXX
9	46	46	2	process wastewater	X066-X200-X027-XXXX
Totals for areas that were used for application	804	802	28		
Totals for areas that were not used for application					
Land application area totals	804	802	28		

B. CROPS AND HARVESTS

1

Field name: 1

11/07/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	1,281.06 *ton*	As-is		62.3	9,200.00	1,200.00	11,000.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.30	336.74	43.92	402.62	1,421.28

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

1

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,992.76 *ton*	As-is		76.4	2,900.00	500.00	3,200.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.47	165.11	28.47	182.20	833.09

10

Field name: 10

11/01/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 23 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/09/2023	431.46 *ton*	As-is		78.8	4,900.00	900.00	8,000.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.76	183.84	33.77	300.15	922.65

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	671.16 *ton*	As-is		63.1	4,800.00	1,000.00	5,900.00		4.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	29.18	280.14	58.36	344.33	926.03

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

11

Field name: 11

11/03/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 32 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/09/2023	557.28 *ton*	As-is		78.8	4,900.00	900.00	800.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	17.42	170.67	31.35	27.86	856.54

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	866.88 *ton*	As-is		56.8	4,700.00	700.00	5,800.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	27.09	254.65	37.93	314.24	1,076.66

12

Field name: 12

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 30 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	409.68 *ton*	As-is		78.8	4,900.00	900.00	800.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	13.66	133.83	24.58	21.85	671.66

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	637.28 *ton*	As-is		69.7	3,800.00	700.00	3,800.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	21.24	161.44	29.74	161.44	708.02

13

Field name: 13

11/04/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 75 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	1,365.30 *ton*	As-is		78.8	4,900.00	900.00	800.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.20	178.40	32.77	29.13	895.35

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	2,123.80 *ton*	As-is		66.3	4,600.00	1,000.00	5,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.32	260.52	56.63	300.16	1,240.58

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2

Field name: 2

11/09/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 74 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	1,358.82 *ton*	As-is		66.0	7,800.00	1,200.00	10,100.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.36	286.45	44.07	370.92	1,423.46

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 74 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	2,113.72 *ton*	As-is		76.4	3,200.00	700.00	3,300.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.56	182.81	39.99	188.52	903.30

3

Field name: 3

11/08/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 14 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/08/2023	296.46 *ton*	As-is		57.1	8,700.00	1,300.00	12,600.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	21.18	368.46	55.06	533.63	1,907.72

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	461.16 *ton*	As-is		69.8	3,400.00	700.00	4,000.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	32.94	223.99	46.12	263.52	1,273.33

4

Field name: 4

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 50 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	1,158.66 *ton*	As-is		60.0	7,300.00	1,200.00	10,100.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	23.17	338.33	55.62	468.10	1,798.24

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,802.36 *ton*	As-is		65.2	4,100.00	700.00	4,900.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	36.05	295.59	50.47	353.26	1,329.71

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5

Field name: 5

10/17/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 118 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/04/2023	2,163.42 *ton*	As-is		70.9	5,200.00	900.00	7,800.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.33	190.67	33.00	286.01	1,120.39

04/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 118 Plant date: 04/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	3,365.32 *ton*	As-is		72.6	3,000.00	600.00	3,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.52	171.12	34.22	188.23	1,015.87

09/04/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 118 Plant date: 09/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/10/2023	1,300.00 *ton*	As-is		65.0	9,400.00	1,500.00	12,200.00		20.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.02	207.12	33.05	268.81	1,580.93

6

Field name: 6

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6

10/18/2022: Wheat, silage, boot stage

Crop: Wheat, silage, boot stage Acres planted: 94 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/04/2023	1,704.78 *ton*	As-is		75.8	4,100.00	700.00	5,800.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.14	148.71	25.39	210.38	798.78

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 94 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/25/2023	2,651.88 *ton*	As-is		72.4	3,000.00	600.00	3,400.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.21	169.27	33.85	191.84	747.49

09/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 94 Plant date: 09/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/09/2023	900.00 *ton*	As-is		65.0	9,500.00	1,600.00	12,300.00		19.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.57	181.91	30.64	235.53	1,286.81

7

Field name: 7

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7

10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	639.54 *ton*	As-is		60.0	8,300.00	1,700.00	16,600.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	9.14	151.66	31.06	303.32	957.48

05/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 05/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/31/2023	1,494.38 *ton*	As-is		68.8	3,200.00	600.00	4,300.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	21.35	136.63	25.62	183.60	666.07

8

Field name: 8

10/24/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 104 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/03/2023	1,922.22 *ton*	As-is		69.6	4,000.00	700.00	6,400.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	18.48	147.86	25.88	236.58	887.77

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8

05/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 104 Plant date: 05/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	2,990.12 *ton*	As-is		66.1	3,100.00	600.00	5,100.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	28.75	178.26	34.50	293.26	955.17

9

Field name: 9

10/25/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 46 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/09/2023	997.38 *ton*	As-is		70.3	3,800.00	800.00	6,500.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	220.80	41.60	132.80	0.00
Total actual harvest content	21.68	164.78	34.69	281.87	1,223.53

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 46 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,351.48 *ton*	As-is		55.7	4,900.00	800.00	5,700.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	268.80	50.40	184.80	0.00
Total actual harvest content	29.38	287.92	47.01	334.93	1,301.53

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/07/2022: Wheat, silage, soft dough

Field name: 1

Crop: Wheat, silage, soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	136.23	22.13	152.50	897.33	3,900,000.00 *gal*
Ag Supply Well	Ground water	1.20	0.00	0.00	228.00	7,650,000.00 *gal*
Application event totals		137.43	22.13	152.50	1,125.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	215.30	37.50	241.84	1,427.00	4,200,000.00 *gal*
Ag Supply Well	Ground water	1.28	0.00	0.00	243.20	8,160,000.00 *gal*
Application event totals		216.59	37.50	241.84	1,670.19	

1 - 05/20/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.24	25.85	193.30	1,078.08	2,880,000.00 *gal*
MID Canal	Surface water	1.62	0.00	0.00	43.61	13,547,520.00 *gal*
Application event totals		135.86	25.85	193.30	1,121.69	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

1 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.29	0.00	0.00	34.78	10,805,760.00 *gal*
Application event totals		1.29	0.00	0.00	34.78	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.33	0.00	0.00	35.82	11,128,320.00 *gal*
Application event totals		1.33	0.00	0.00	35.82	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.27	0.00	0.00	34.26	10,644,480.00 *gal*
Application event totals		1.27	0.00	0.00	34.26	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.00	0.00	0.00	26.99	8,386,560.00 *gal*
Application event totals		1.00	0.00	0.00	26.99	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.11	15.59	118.45	615.15	2,400,000.00 *gal*
MID Canal	Surface water	1.02	0.00	0.00	27.51	8,547,840.00 *gal*
Application event totals		77.13	15.59	118.45	642.66	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.88	0.00	0.00	23.88	7,418,880.00 *gal*
Application event totals		0.88	0.00	0.00	23.88	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

1 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.77	0.00	0.00	20.77	6,451,200.00 *gal*
Application event totals		0.77	0.00	0.00	20.77	

10 - 11/01/2022: Wheat, silage, soft dough

Field name: 10

Crop: Wheat, silage, soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.81	18.65	128.53	756.27	1,080,000.00 *gal*
Ag Supply Well	Ground water	1.94	0.00	0.00	309.54	3,468,000.00 *gal*
Application event totals		116.75	18.65	128.53	1,065.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.69	21.20	136.69	806.56	780,000.00 *gal*
Ag Supply Well	Ground water	1.71	0.00	0.00	273.12	3,060,000.00 *gal*
Application event totals		123.40	21.20	136.69	1,079.68	

10 - 05/10/2023: Corn, silage

Field name: 10

Crop: Corn, silage Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	135.66	40.82	152.47	0.00	90.00 *ton*
Application event totals		135.66	40.82	152.47	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.23	0.00	0.00	33.18	3,386,880.00 *gal*
Application event totals		1.23	0.00	0.00	33.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.17	0.00	0.00	31.60	3,225,600.00 *gal*
Application event totals		1.17	0.00	0.00	31.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	153.21	29.51	220.61	1,230.42	1,080,000.00 *gal*
MID Canal	Surface water	1.23	0.00	0.00	33.18	3,386,880.00 *gal*
Application event totals		154.44	29.51	220.61	1,263.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.11	0.00	0.00	30.02	3,064,320.00 *gal*
Application event totals		1.11	0.00	0.00	30.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.99	0.00	0.00	26.86	2,741,760.00 *gal*
Application event totals		0.99	0.00	0.00	26.86	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.07	16.61	126.18	655.26	840,000.00 *gal*
MID Canal	Surface water	0.94	0.00	0.00	25.28	2,580,480.00 *gal*
Application event totals		82.01	16.61	126.18	680.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.88	0.00	0.00	23.70	2,419,200.00 *gal*
Application event totals		0.88	0.00	0.00	23.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.11	0.00	0.00	30.02	3,064,320.00 *gal*
Application event totals		1.11	0.00	0.00	30.02	

11 - 11/03/2022: Wheat, silage, soft dough

Field name: 11

Crop: Wheat, silage, soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.35	11.92	82.11	483.18	960,000.00 *gal*
Ag Supply Well	Ground water	1.88	0.00	0.00	301.00	4,692,000.00 *gal*
Application event totals		75.24	11.92	82.11	784.18	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

11 - 11/03/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.29	24.61	158.71	936.47	1,260,000.00 *gal*
Ag Supply Well	Ground water	2.09	0.00	0.00	333.72	5,202,000.00 *gal*
Application event totals		143.38	24.61	158.71	1,270.19	

11 - 05/11/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.30	32.99	246.66	1,375.67	1,680,000.00 *gal*
MID Canal	Surface water	1.18	0.00	0.00	31.80	4,515,840.00 *gal*
Application event totals		172.48	32.99	246.66	1,407.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.30	0.00	0.00	35.20	4,999,680.00 *gal*
Application event totals		1.30	0.00	0.00	35.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.09	0.00	0.00	29.53	4,193,280.00 *gal*
Application event totals		1.09	0.00	0.00	29.53	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

11 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.14	0.00	0.00	30.66	4,354,560.00 *gal*
Application event totals		1.14	0.00	0.00	30.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.01	0.00	0.00	27.25	3,870,720.00 *gal*
Application event totals		1.01	0.00	0.00	27.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.57	18.76	142.51	740.10	1,320,000.00 *gal*
MID Canal	Surface water	0.93	0.00	0.00	24.98	3,548,160.00 *gal*
Application event totals		92.49	18.76	142.51	765.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.97	0.00	0.00	26.12	3,709,440.00 *gal*
Application event totals		0.97	0.00	0.00	26.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.22	0.00	0.00	32.93	4,677,120.00 *gal*
Application event totals		1.22	0.00	0.00	32.93	

12 - 11/02/2022: Wheat, silage, soft dough

Field name: 12

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12 - 11/02/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.90	7.94	54.74	322.12	600,000.00 *gal*
Ag Supply Well	Ground water	1.75	0.00	0.00	279.19	4,080,000.00 *gal*
Application event totals		50.65	7.94	54.74	601.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.83	20.00	128.98	761.06	960,000.00 *gal*
Ag Supply Well	Ground water	1.57	0.00	0.00	251.27	3,672,000.00 *gal*
Application event totals		116.40	20.00	128.98	1,012.34	

12 - 05/12/2023: Corn, silage

Field name: 12

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.52	25.14	187.93	1,048.13	1,200,000.00 *gal*
MID Canal	Surface water	1.35	0.00	0.00	36.34	4,838,400.00 *gal*
Application event totals		131.86	25.14	187.93	1,084.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.30	0.00	0.00	35.13	4,677,120.00 *gal*
Application event totals		1.30	0.00	0.00	35.13	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.08	0.00	0.00	29.07	3,870,720.00 *gal*
Application event totals		1.08	0.00	0.00	29.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.90	0.00	0.00	24.23	3,225,600.00 *gal*
Application event totals		0.90	0.00	0.00	24.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.03	14.55	110.55	574.14	960,000.00 *gal*
MID Canal	Surface water	1.17	0.00	0.00	31.49	4,193,280.00 *gal*
Application event totals		72.20	14.55	110.55	605.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.08	0.00	0.00	29.07	3,870,720.00 *gal*
Application event totals		1.08	0.00	0.00	29.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.94	0.00	0.00	25.44	3,386,880.00 *gal*
Application event totals		0.94	0.00	0.00	25.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.26	0.00	0.00	33.92	4,515,840.00 *gal*
Application event totals		1.26	0.00	0.00	33.92	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 11/04/2022: Wheat, silage, soft dough

Field name: 13

Crop: Wheat, silage, soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.80	15.89	109.49	644.23	3,000,000.00 *gal*
Ag Supply Well	Ground water	1.57	0.00	0.00	251.27	9,180,000.00 *gal*
Application event totals		99.38	15.89	109.49	895.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.92	23.50	151.55	894.25	2,820,000.00 *gal*
Ag Supply Well	Ground water	1.52	0.00	0.00	242.90	8,874,000.00 *gal*
Application event totals		136.44	23.50	151.55	1,137.15	

13 - 05/15/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	126.38	14.58	176.38	0.00	1,200.00 *ton*
Application event totals		126.38	14.58	176.38	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.44	0.00	0.00	38.76	12,902,400.00 *gal*
Application event totals		1.44	0.00	0.00	38.76	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.36	0.00	0.00	36.82	12,257,280.00 *gal*
Application event totals		1.36	0.00	0.00	36.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.57	27.65	206.72	1,152.95	3,300,000.00 *gal*
MID Canal	Surface water	1.33	0.00	0.00	35.85	11,934,720.00 *gal*
Application event totals		144.90	27.65	206.72	1,188.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.18	0.00	0.00	31.98	10,644,480.00 *gal*
Application event totals		1.18	0.00	0.00	31.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.97	0.00	0.00	26.16	8,709,120.00 *gal*
Application event totals		0.97	0.00	0.00	26.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.81	14.92	113.32	588.49	2,460,000.00 *gal*
MID Canal	Surface water	1.09	0.00	0.00	29.56	9,838,080.00 *gal*
Application event totals		73.90	14.92	113.32	618.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.93	0.00	0.00	25.19	8,386,560.00 *gal*
Application event totals		0.93	0.00	0.00	25.19	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.81	0.00	0.00	21.80	7,257,600.00 *gal*
Application event totals		0.81	0.00	0.00	21.80	

2 - 11/09/2022: Wheat, silage, soft dough

Field name: 2

Crop: Wheat, silage, soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	129.12	18.45	29.51	0.00	1,000.00 *ton*
Application event totals		129.12	18.45	29.51	0.00	

10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.06	17.39	119.84	705.18	3,240,000.00 *gal*
Ag Supply Well	Ground water	1.28	0.00	0.00	241.55	8,568,000.00 *gal*
Application event totals		108.33	17.39	119.84	946.73	

01/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.56	19.26	124.19	732.78	2,280,000.00 *gal*
Ag Supply Well	Ground water	1.18	0.00	0.00	224.30	7,956,000.00 *gal*
Application event totals		111.74	19.26	124.19	957.08	

2 - 05/19/2023: Corn, silage

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2 - 05/19/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	166.67	32.10	239.99	1,338.49	3,780,000.00 *gal*
MID Canal	Surface water	1.51	0.00	0.00	40.76	13,386,240.00 *gal*
Application event totals		168.18	32.10	239.99	1,379.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.29	0.00	0.00	34.87	11,450,880.00 *gal*
Application event totals		1.29	0.00	0.00	34.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.13	0.00	0.00	30.45	9,999,360.00 *gal*
Application event totals		1.13	0.00	0.00	30.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.93	0.00	0.00	25.04	8,225,280.00 *gal*
Application event totals		0.93	0.00	0.00	25.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.99	14.75	112.05	581.89	2,400,000.00 *gal*
MID Canal	Surface water	0.95	0.00	0.00	25.54	8,386,560.00 *gal*
Application event totals		72.94	14.75	112.05	607.43	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.89	0.00	0.00	24.06	7,902,720.00 *gal*
Application event totals		0.89	0.00	0.00	24.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.96	0.00	0.00	26.03	8,547,840.00 *gal*
Application event totals		0.96	0.00	0.00	26.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.84	0.00	0.00	22.59	7,418,880.00 *gal*
Application event totals		0.84	0.00	0.00	22.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.78	0.00	0.00	21.12	6,935,040.00 *gal*
Application event totals		0.78	0.00	0.00	21.12	

3 - 11/08/2022: Wheat, silage, soft dough

Field name: 3

Crop: Wheat, silage, soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	143.33	20.48	32.76	0.00	210.00 *ton*
Application event totals		143.33	20.48	32.76	0.00	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3 - 11/08/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.27	18.73	129.04	759.28	660,000.00 *gal*
Ag Supply Well	Ground water	1.61	0.00	0.00	304.00	2,040,000.00 *gal*
Application event totals		116.87	18.73	129.04	1,063.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.41	24.11	155.47	917.35	540,000.00 *gal*
Ag Supply Well	Ground water	1.44	0.00	0.00	273.60	1,836,000.00 *gal*
Application event totals		139.85	24.11	155.47	1,190.95	

3 - 05/17/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	181.79	35.01	261.76	1,459.90	780,000.00 *gal*
MID Canal	Surface water	1.63	0.00	0.00	44.13	2,741,760.00 *gal*
Application event totals		183.42	35.01	261.76	1,504.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.35	0.00	0.00	36.34	2,257,920.00 *gal*
Application event totals		1.35	0.00	0.00	36.34	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.15	0.00	0.00	31.15	1,935,360.00 *gal*
Application event totals		1.15	0.00	0.00	31.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.06	0.00	0.00	28.55	1,774,080.00 *gal*
Application event totals		1.06	0.00	0.00	28.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.65	21.44	162.87	845.83	660,000.00 *gal*
MID Canal	Surface water	1.25	0.00	0.00	33.74	2,096,640.00 *gal*
Application event totals		105.90	21.44	162.87	879.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.87	0.00	0.00	23.36	1,451,520.00 *gal*
Application event totals		0.87	0.00	0.00	23.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.96	0.00	0.00	25.96	1,612,800.00 *gal*
Application event totals		0.96	0.00	0.00	25.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.77	0.00	0.00	20.77	1,290,240.00 *gal*
Application event totals		0.77	0.00	0.00	20.77	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4 - 10/27/2022: Wheat, silage, soft dough

Field name: 4

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	140.84	22.88	157.66	927.70	2,880,000.00 *gal*
Ag Supply Well	Ground water	1.66	0.00	0.00	254.05	6,264,000.00 *gal*
Application event totals		142.50	22.88	157.66	1,181.74	

01/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	219.61	38.25	246.67	1,455.53	3,060,000.00 *gal*
Ag Supply Well	Ground water	1.75	0.00	0.00	267.19	6,588,000.00 *gal*
Application event totals		221.36	38.25	246.67	1,722.72	

4 - 05/18/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	166.41	50.07	187.03	0.00	240.00 *ton*
Application event totals		166.41	50.07	187.03	0.00	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.83	0.00	0.00	280.33	6,912,000.00 *gal*
Application event totals		1.83	0.00	0.00	280.33	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.69	0.00	0.00	258.43	6,372,000.00 *gal*
Application event totals		1.69	0.00	0.00	258.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.13	25.64	191.69	1,069.09	2,040,000.00 *gal*
Ag Supply Well	Ground water	1.75	0.00	0.00	267.19	6,588,000.00 *gal*
Application event totals		134.87	25.64	191.69	1,336.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.78	0.00	0.00	271.57	6,696,000.00 *gal*
Application event totals		1.78	0.00	0.00	271.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.59	13.64	103.64	538.25	1,500,000.00 *gal*
Ag Supply Well	Ground water	1.86	0.00	0.00	284.71	7,020,000.00 *gal*
Application event totals		68.46	13.64	103.64	822.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.95	0.00	0.00	297.85	7,344,000.00 *gal*
Application event totals		1.95	0.00	0.00	297.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.98	0.00	0.00	302.23	7,452,000.00 *gal*
Application event totals		1.98	0.00	0.00	302.23	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 10/17/2022: Wheat, silage, soft dough

Field name: 5

Crop: Wheat, silage, soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	136.04	19.43	31.09	0.00	1,680.00 *ton*
Application event totals		136.04	19.43	31.09	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.75	0.00	0.00	267.26	15,552,000.00 *gal*
Application event totals		1.75	0.00	0.00	267.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.95	20.02	129.12	761.87	3,780,000.00 *gal*
Ag Supply Well	Ground water	1.86	0.00	0.00	283.97	16,524,000.00 *gal*
Application event totals		116.81	20.02	129.12	1,045.84	

5 - 04/26/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 04/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	132.73	25.56	191.11	1,065.90	4,800,000.00 *gal*
Ag Supply Well	Ground water	2.00	0.00	0.00	306.24	17,820,000.00 *gal*
Application event totals		134.73	25.56	191.11	1,372.13	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 04/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	2.02	0.00	0.00	309.02	17,982,000.00 *gal*
Application event totals		2.02	0.00	0.00	309.02	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	2.09	0.00	0.00	320.16	18,630,000.00 *gal*
Application event totals		2.09	0.00	0.00	320.16	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.79	0.00	0.00	272.83	15,876,000.00 *gal*
Application event totals		1.79	0.00	0.00	272.83	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.71	0.00	0.00	261.69	15,228,000.00 *gal*
Application event totals		1.71	0.00	0.00	261.69	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.27	16.65	126.48	656.85	4,320,000.00 *gal*
Ag Supply Well	Ground water	1.68	0.00	0.00	256.13	14,904,000.00 *gal*
Application event totals		82.94	16.65	126.48	912.98	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.62	0.00	0.00	247.77	14,418,000.00 *gal*
Application event totals		1.62	0.00	0.00	247.77	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 09/04/2023: Sudangrass, silage

Field name: 5

Crop: Sudangrass, silage

Plant date: 09/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.10	21.74	165.13	857.55	5,640,000.00 *gal*
Ag Supply Well	Ground water	1.71	0.00	0.00	261.69	15,228,000.00 *gal*
Application event totals		107.81	21.74	165.13	1,119.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.30	5.61	153.57	878.60	5,220,000.00 *gal*
Ag Supply Well	Ground water	1.58	0.00	0.00	242.21	14,094,000.00 *gal*
Application event totals		100.89	5.61	153.57	1,120.81	

6 - 10/18/2022: Wheat, silage, boot stage

Field name: 6

Crop: Wheat, silage, boot stage

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	199.30	24.16	58.67	0.00	500.00 *ton*
Application event totals		199.30	24.16	58.67	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.10	0.00	0.00	284.88	12,342,000.00 *gal*
Application event totals		1.10	0.00	0.00	284.88	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 10/18/2022: Wheat, silage, boot stage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.17	0.00	0.00	303.71	13,158,000.00 *gal*
Application event totals		1.17	0.00	0.00	303.71	

6 - 04/28/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.88	23.66	176.93	986.81	3,540,000.00 *gal*
Ag Supply Well	Ground water	1.08	0.00	0.00	280.17	12,138,000.00 *gal*
Application event totals		123.96	23.66	176.93	1,266.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.21	0.00	0.00	315.48	13,668,000.00 *gal*
Application event totals		1.21	0.00	0.00	315.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.28	0.00	0.00	331.96	14,382,000.00 *gal*
Application event totals		1.28	0.00	0.00	331.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.10	0.00	0.00	287.23	12,444,000.00 *gal*
Application event totals		1.10	0.00	0.00	287.23	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.68	18.58	141.13	732.94	3,840,000.00 *gal*
Ag Supply Well	Ground water	1.12	0.00	0.00	291.94	12,648,000.00 *gal*
Application event totals		91.80	18.58	141.13	1,024.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.23	0.00	0.00	320.19	13,872,000.00 *gal*
Application event totals		1.23	0.00	0.00	320.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	0.98	0.00	0.00	254.27	11,016,000.00 *gal*
Application event totals		0.98	0.00	0.00	254.27	

6 - 09/01/2023: Sudangrass, silage

Field name: 6

Crop: Sudangrass, silage

Plant date: 09/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.77	23.51	178.62	927.63	4,860,000.00 *gal*
Ag Supply Well	Ground water	0.91	0.00	0.00	237.79	10,302,000.00 *gal*
Application event totals		115.68	23.51	178.62	1,165.42	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 09/01/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.70	5.18	141.82	811.35	3,840,000.00 *gal*
Ag Supply Well	Ground water	0.85	0.00	0.00	221.31	9,588,000.00 *gal*
Application event totals		92.55	5.18	141.82	1,032.66	

7 - 10/21/2022: Wheat, silage, soft dough

Field name: 7

Crop: Wheat, silage, soft dough Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	167.90	23.99	38.38	0.00	1,230.00 *ton*
Application event totals		167.90	23.99	38.38	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.14	0.00	0.00	282.33	9,588,000.00 *gal*
Application event totals		1.14	0.00	0.00	282.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.11	0.00	0.00	273.32	9,282,000.00 *gal*
Application event totals		1.11	0.00	0.00	273.32	

7 - 05/03/2023: Corn, silage

Field name: 7

Crop: Corn, silage Plant date: 05/03/2023

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7 - 05/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.06	23.70	177.19	988.24	2,640,000.00 *gal*
Ag Supply Well	Ground water	1.31	0.00	0.00	324.38	11,016,000.00 *gal*
Application event totals		124.37	23.70	177.19	1,312.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.11	0.00	0.00	273.32	9,282,000.00 *gal*
Application event totals		1.11	0.00	0.00	273.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.12	0.00	0.00	276.32	9,384,000.00 *gal*
Application event totals		1.12	0.00	0.00	276.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.06	0.00	0.00	261.30	8,874,000.00 *gal*
Application event totals		1.06	0.00	0.00	261.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.03	0.00	0.00	255.30	8,670,000.00 *gal*
Application event totals		1.03	0.00	0.00	255.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.89	12.47	94.76	492.12	1,920,000.00 *gal*
Ag Supply Well	Ground water	1.14	0.00	0.00	282.33	9,588,000.00 *gal*
Application event totals		62.03	12.47	94.76	774.44	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7 - 05/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.20	0.00	0.00	297.34	10,098,000.00 *gal*
Application event totals		1.20	0.00	0.00	297.34	

8 - 10/24/2022: Wheat, silage, soft dough

Field name: 8

Crop: Wheat, silage, soft dough Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.64	13.75	94.75	557.51	3,600,000.00 *gal*
Ag Supply Well	Ground water	1.01	0.00	0.00	248.65	12,546,000.00 *gal*
Application event totals		85.64	13.75	94.75	806.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.79	19.47	125.57	740.94	3,240,000.00 *gal*
Ag Supply Well	Ground water	0.93	0.00	0.00	230.46	11,628,000.00 *gal*
Application event totals		112.72	19.47	125.57	971.40	

8 - 05/05/2023: Corn, silage

Field name: 8

Crop: Corn, silage Plant date: 05/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8 - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	146.83	28.28	211.42	1,179.15	4,680,000.00 *gal*
Ag Supply Well	Ground water	1.05	0.00	0.00	258.76	13,056,000.00 *gal*
Application event totals		147.88	28.28	211.42	1,437.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.19	0.00	0.00	293.13	14,790,000.00 *gal*
Application event totals		1.19	0.00	0.00	293.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.23	0.00	0.00	303.24	15,300,000.00 *gal*
Application event totals		1.23	0.00	0.00	303.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.07	0.00	0.00	264.83	13,362,000.00 *gal*
Application event totals		1.07	0.00	0.00	264.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.40	16.27	123.58	641.76	3,720,000.00 *gal*
Ag Supply Well	Ground water	1.00	0.00	0.00	246.63	12,444,000.00 *gal*
Application event totals		80.40	16.27	123.58	888.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.06	0.00	0.00	262.80	13,260,000.00 *gal*
Application event totals		1.06	0.00	0.00	262.80	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8 - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.06	0.00	0.00	260.78	13,158,000.00 *gal*
Application event totals		1.06	0.00	0.00	260.78	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.19	0.00	0.00	293.13	14,790,000.00 *gal*
Application event totals		1.19	0.00	0.00	293.13	

9 - 10/25/2022: Wheat, silage, soft dough

Field name: 9

Crop: Wheat, silage, soft dough Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.62	18.13	124.96	735.27	2,100,000.00 *gal*
Ag Supply Well	Ground water	1.20	0.00	0.00	297.08	6,630,000.00 *gal*
Application event totals		112.83	18.13	124.96	1,032.35	

01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.29	17.12	110.40	651.45	1,260,000.00 *gal*
Ag Supply Well	Ground water	1.13	0.00	0.00	278.80	6,222,000.00 *gal*
Application event totals		99.42	17.12	110.40	930.26	

9 - 05/08/2023: Corn, silage

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

9 - 05/08/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	178.75	34.42	257.38	1,435.49	2,520,000.00 *gal*
Ag Supply Well	Ground water	1.77	0.00	0.00	282.22	6,324,000.00 *gal*
Application event totals		180.52	34.42	257.38	1,717.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.45	0.00	0.00	232.15	5,202,000.00 *gal*
Application event totals		1.45	0.00	0.00	232.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.99	0.00	0.00	318.64	7,140,000.00 *gal*
Application event totals		1.99	0.00	0.00	318.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.82	0.00	0.00	291.33	6,528,000.00 *gal*
Application event totals		1.82	0.00	0.00	291.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.74	0.00	0.00	277.67	6,222,000.00 *gal*
Application event totals		1.74	0.00	0.00	277.67	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

9 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.19	27.29	207.29	1,076.51	2,760,000.00 *gal*
Ag Supply Well	Ground water	1.60	0.00	0.00	254.91	5,712,000.00 *gal*
Application event totals		134.78	27.29	207.29	1,331.42	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.82	0.00	0.00	291.33	6,528,000.00 *gal*
Application event totals		1.82	0.00	0.00	291.33	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

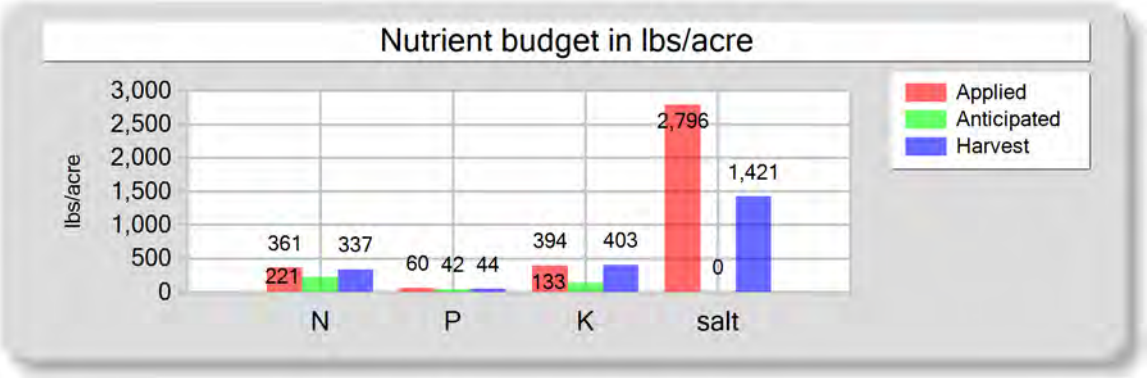
Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Supply Well	Ground water	1.80	0.00	0.00	286.78	6,426,000.00 *gal*
Application event totals		1.80	0.00	0.00	286.78	

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

B. NUTRIENT BUDGET

1 - 11/07/2022: Wheat, silage, soft dough

Field name: 1 Crop: Wheat, silage, soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	351.53	59.63	394.34	2,324.32
Fresh water	2.49	0.00	0.00	471.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	361.02	59.63	394.34	2,795.52
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	336.74	43.92	402.62	1,421.28
Nutrient balance	24.28	15.71	-8.28	1,374.23
Applied to removed ratio	1.07	1.36	0.98	1.97

Fresh water applied	
15,810,000.00	*gallons*
582.23	*acre-inches*
8.32	*inches/acre*

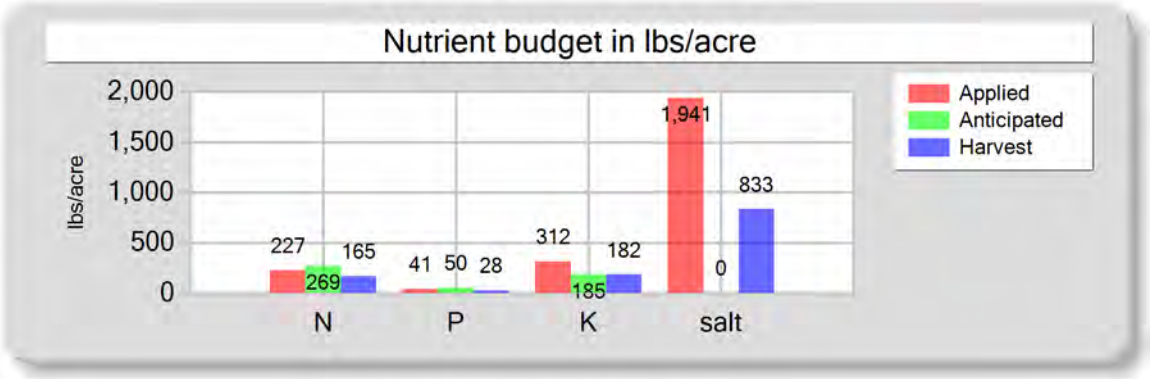
Process wastewater applied	
8,100,000.00	*gallons*
298.30	*acre-inches*
4.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

1 - 05/20/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.35	41.45	311.75	1,693.22
Fresh water	9.17	0.00	0.00	247.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.52	41.45	311.75	1,940.85
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	165.11	28.47	182.20	833.09
Nutrient balance	61.41	12.98	129.55	1,107.76
Applied to removed ratio	1.37	1.46	1.71	2.33

Fresh water applied	
76,930,560.00	*gallons*
2,833.09	*acre-inches*
40.47	*inches/acre*

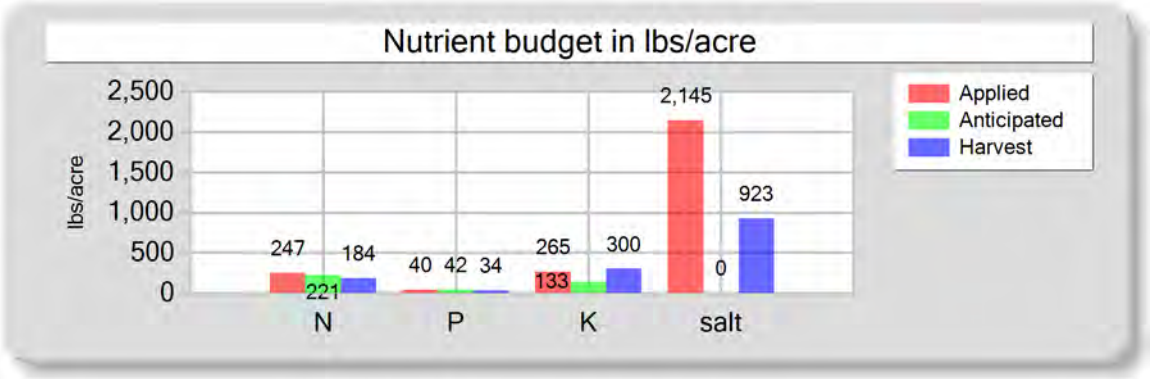
Process wastewater applied	
5,280,000.00	*gallons*
194.44	*acre-inches*
2.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10 - 11/01/2022: Wheat, silage, soft dough

Field name: 10 Crop: Wheat, silage, soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.50	39.85	265.22	1,562.84
Fresh water	3.65	0.00	0.00	582.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.15	39.85	265.22	2,145.50
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	183.84	33.77	300.15	922.65
Nutrient balance	63.31	6.08	-34.93	1,222.85
Applied to removed ratio	1.34	1.18	0.88	2.33

Fresh water applied	
6,528,000.00	*gallons*
240.40	*acre-inches*
10.45	*inches/acre*

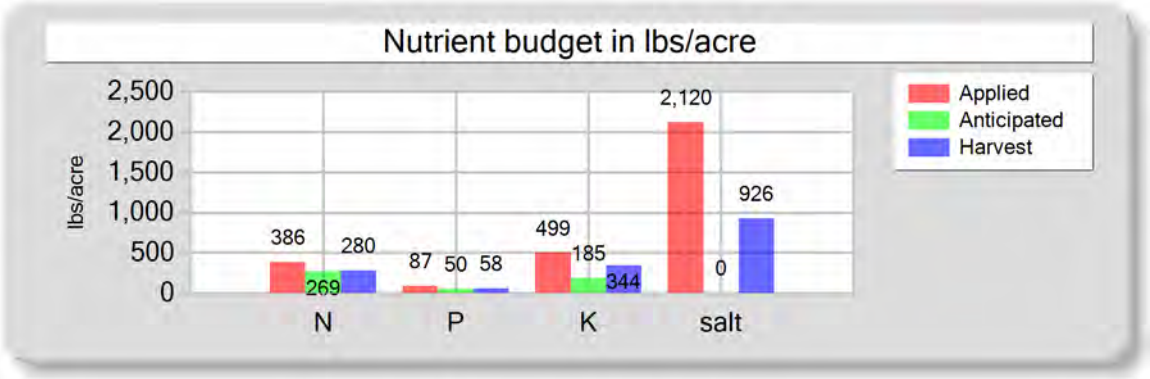
Process wastewater applied	
1,860,000.00	*gallons*
68.50	*acre-inches*
2.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10 - 05/10/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	135.66	40.82	152.47	0.00
Process wastewater	234.28	46.12	346.79	1,885.68
Fresh water	8.66	0.00	0.00	233.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	385.60	86.93	499.26	2,119.51
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	280.14	58.36	344.33	926.03
Nutrient balance	105.47	28.57	154.92	1,193.49
Applied to removed ratio	1.38	1.49	1.45	2.29

Fresh water applied	
23,869,440.00	*gallons*
879.03	*acre-inches*
38.22	*inches/acre*

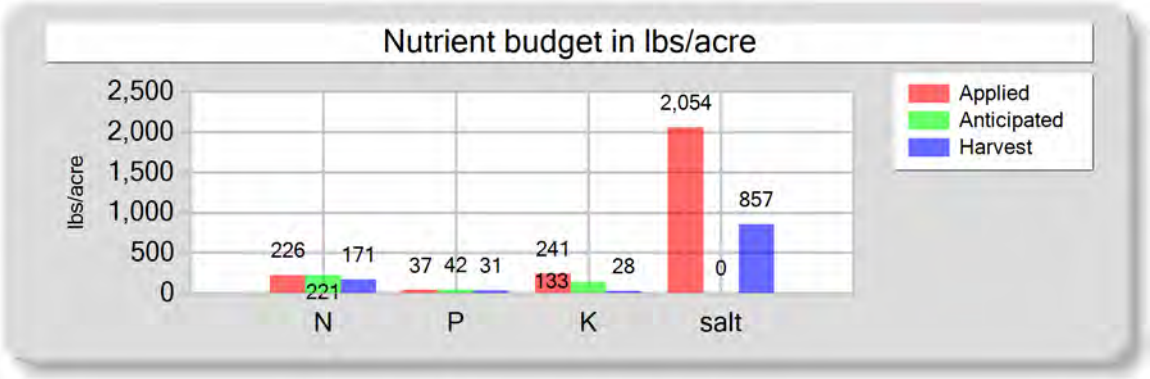
Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
3.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

11 - 11/03/2022: Wheat, silage, soft dough

Field name: 11 Crop: Wheat, silage, soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.64	36.53	240.82	1,419.64
Fresh water	3.97	0.00	0.00	634.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.62	36.53	240.82	2,054.36
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	170.67	31.35	27.86	856.54
Nutrient balance	54.95	5.18	212.96	1,197.82
Applied to removed ratio	1.32	1.17	8.64	2.40

Fresh water applied
9,894,000.00 *gallons*
364.36 *acre-inches*
11.39 *inches/acre*

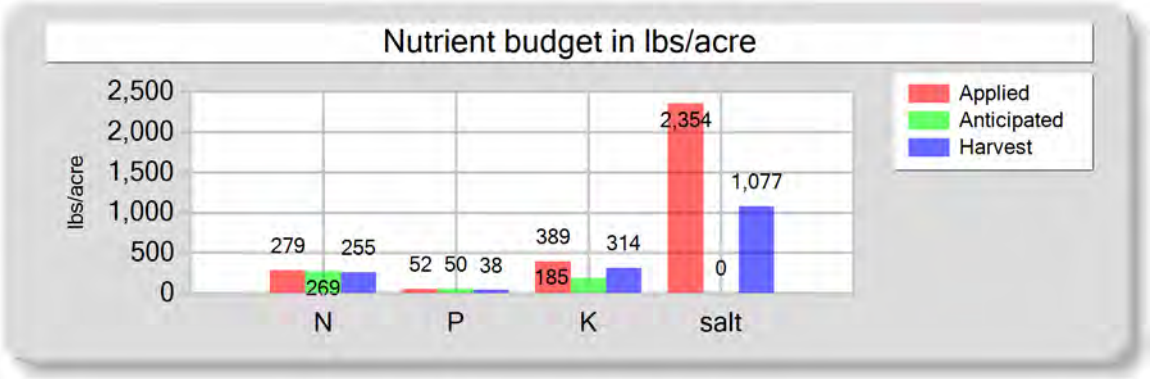
Process wastewater applied
2,220,000.00 *gallons*
81.76 *acre-inches*
2.55 *inches/acre*

Total harvests for the crop
1 *harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

11 - 05/11/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.87	51.75	389.17	2,115.77
Fresh water	8.83	0.00	0.00	238.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.70	51.75	389.17	2,354.24
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	254.65	37.93	314.24	1,076.66
Nutrient balance	24.05	13.82	74.93	1,277.58
Applied to removed ratio	1.09	1.36	1.24	2.19

Fresh water applied	
33,868,800.00	*gallons*
1,247.27	*acre-inches*
38.98	*inches/acre*

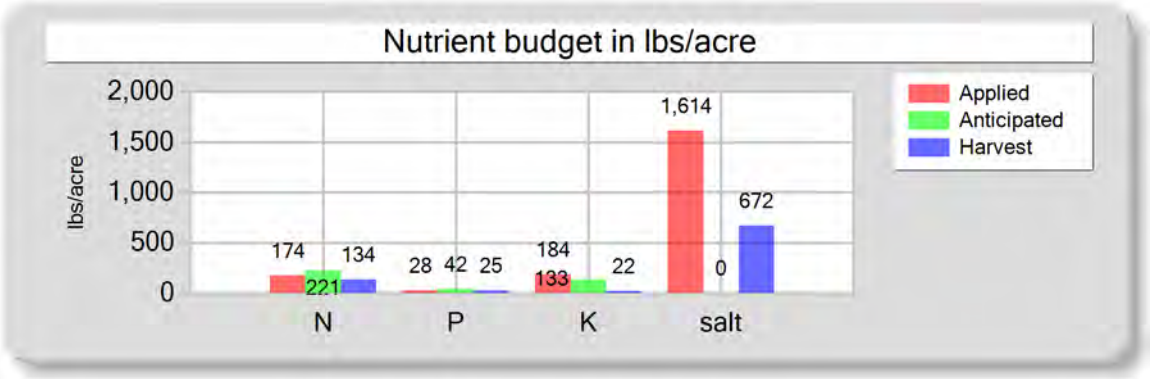
Process wastewater applied	
3,000,000.00	*gallons*
110.48	*acre-inches*
3.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12 - 11/02/2022: Wheat, silage, soft dough

Field name: 12 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.73	27.95	183.72	1,083.18
Fresh water	3.32	0.00	0.00	530.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.05	27.95	183.72	1,613.64
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	133.83	24.58	21.85	671.66
Nutrient balance	40.22	3.36	161.87	941.99
Applied to removed ratio	1.30	1.14	8.41	2.40

Fresh water applied	
7,752,000.00	*gallons*
285.48	*acre-inches*
9.52	*inches/acre*

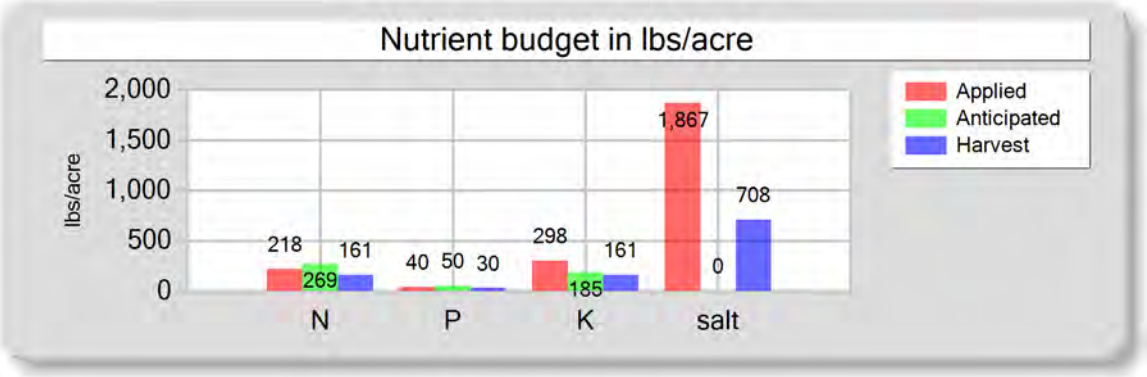
Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12 - 05/12/2023: Corn, silage

Field name: 12 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.55	39.69	298.48	1,622.27
Fresh water	9.06	0.00	0.00	244.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.61	39.69	298.48	1,866.95
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	161.44	29.74	161.44	708.02
Nutrient balance	56.17	9.95	137.04	1,158.93
Applied to removed ratio	1.35	1.33	1.85	2.64

Fresh water applied	
32,578,560.00	*gallons*
1,199.76	*acre-inches*
39.99	*inches/acre*

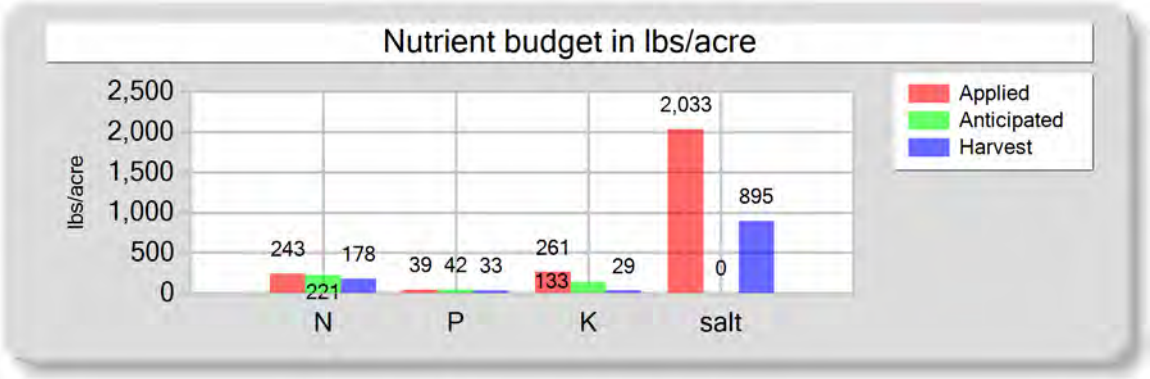
Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 11/04/2022: Wheat, silage, soft dough

Field name: 13 Crop: Wheat, silage, soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.73	39.39	261.04	1,538.48
Fresh water	3.09	0.00	0.00	494.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.82	39.39	261.04	2,032.65
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	178.40	32.77	29.13	895.35
Nutrient balance	64.42	6.62	231.91	1,137.31
Applied to removed ratio	1.36	1.20	8.96	2.27

Fresh water applied	
18,054,000.00	*gallons*
664.87	*acre-inches*
8.86	*inches/acre*

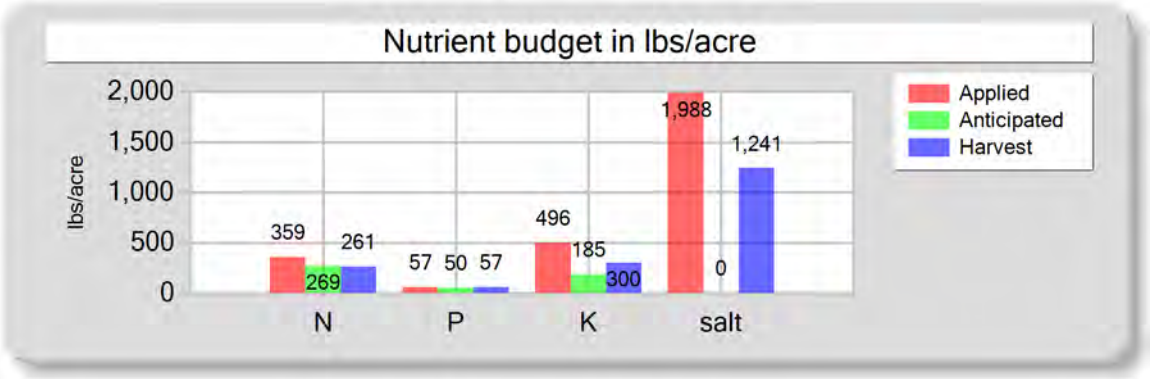
Process wastewater applied	
5,820,000.00	*gallons*
214.33	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 05/15/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	126.38	14.58	176.38	0.00
Process wastewater	216.38	42.57	320.04	1,741.43
Fresh water	9.12	0.00	0.00	246.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	358.87	57.15	496.42	1,987.57
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	260.52	56.63	300.16	1,240.58
Nutrient balance	98.35	0.51	196.25	746.99
Applied to removed ratio	1.38	1.01	1.65	1.60

Fresh water applied	
81,930,240.00	*gallons*
3,017.21	*acre-inches*
40.23	*inches/acre*

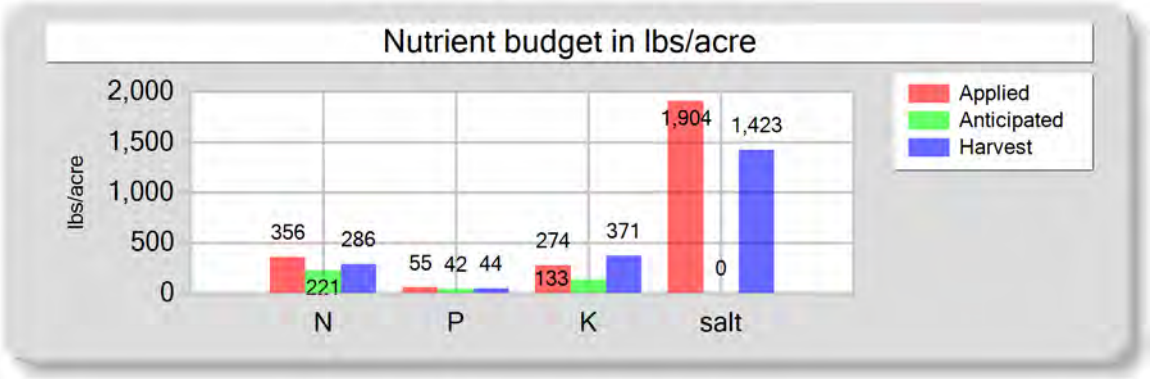
Process wastewater applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
2.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2 - 11/09/2022: Wheat, silage, soft dough

Field name: 2 Crop: Wheat, silage, soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.12	18.45	29.51	0.00
Process wastewater	217.62	36.65	244.03	1,437.96
Fresh water	2.46	0.00	0.00	465.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.20	55.10	273.54	1,903.81
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	286.45	44.07	370.92	1,423.46
Nutrient balance	69.74	11.03	-97.38	480.35
Applied to removed ratio	1.24	1.25	0.74	1.34

Fresh water applied	
16,524,000.00	*gallons*
608.52	*acre-inches*
8.22	*inches/acre*

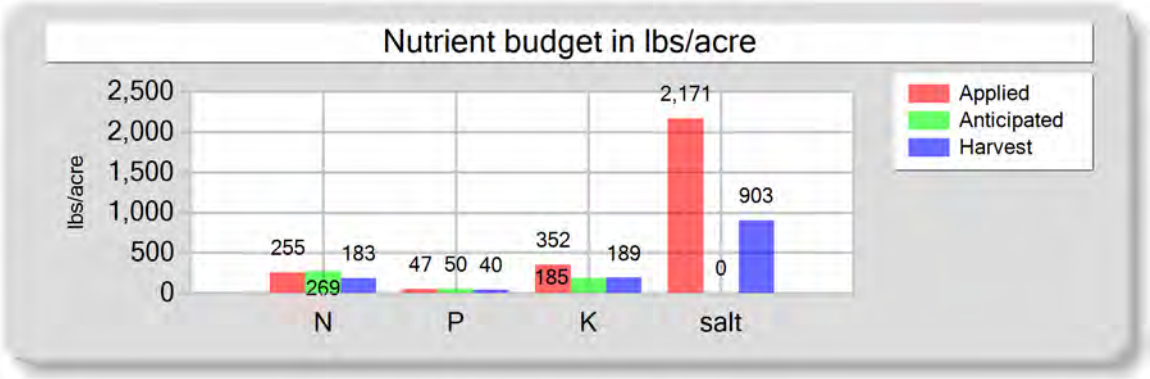
Process wastewater applied	
5,520,000.00	*gallons*
203.28	*acre-inches*
2.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2 - 05/19/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.66	46.85	352.04	1,920.39
Fresh water	9.28	0.00	0.00	250.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.94	46.85	352.04	2,170.83
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	182.81	39.99	188.52	903.30
Nutrient balance	72.13	6.86	163.52	1,267.53
Applied to removed ratio	1.39	1.17	1.87	2.40

Fresh water applied	
82,252,800.00	*gallons*
3,029.09	*acre-inches*
40.93	*inches/acre*

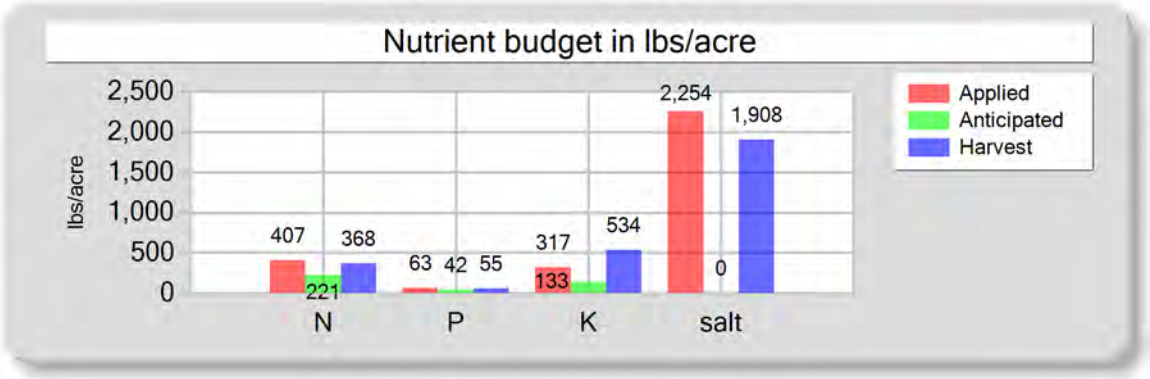
Process wastewater applied	
6,180,000.00	*gallons*
227.59	*acre-inches*
3.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3 - 11/08/2022: Wheat, silage, soft dough

Field name: 3 Crop: Wheat, silage, soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.33	20.48	32.76	0.00
Process wastewater	253.68	42.83	284.50	1,676.63
Fresh water	3.05	0.00	0.00	577.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	407.05	63.31	317.26	2,254.22
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	368.46	55.06	533.63	1,907.72
Nutrient balance	38.59	8.25	-216.36	346.50
Applied to removed ratio	1.10	1.15	0.59	1.18

Fresh water applied	
3,876,000.00	*gallons*
142.74	*acre-inches*
10.20	*inches/acre*

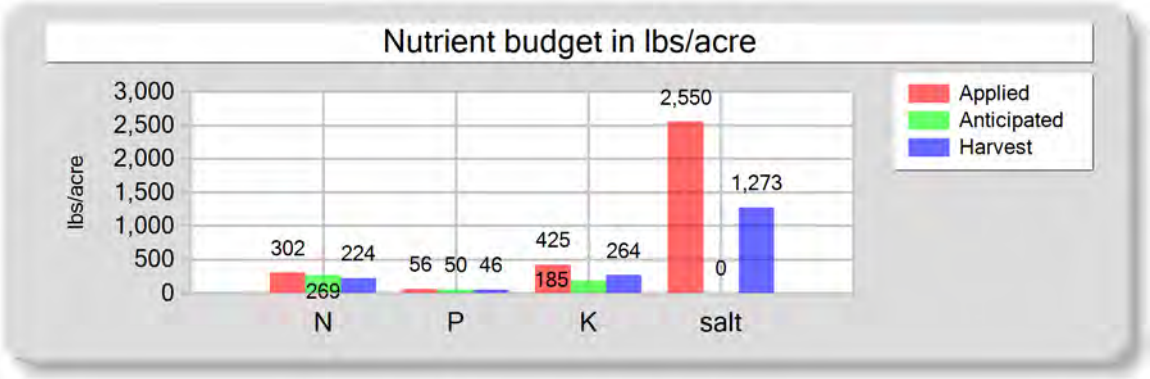
Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
3.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3 - 05/17/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.44	56.45	424.63	2,305.72
Fresh water	9.04	0.00	0.00	243.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.47	56.45	424.63	2,549.71
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	223.99	46.12	263.52	1,273.33
Nutrient balance	78.48	10.33	161.11	1,276.38
Applied to removed ratio	1.35	1.22	1.61	2.00

Fresh water applied	
15,160,320.00	*gallons*
558.30	*acre-inches*
39.88	*inches/acre*

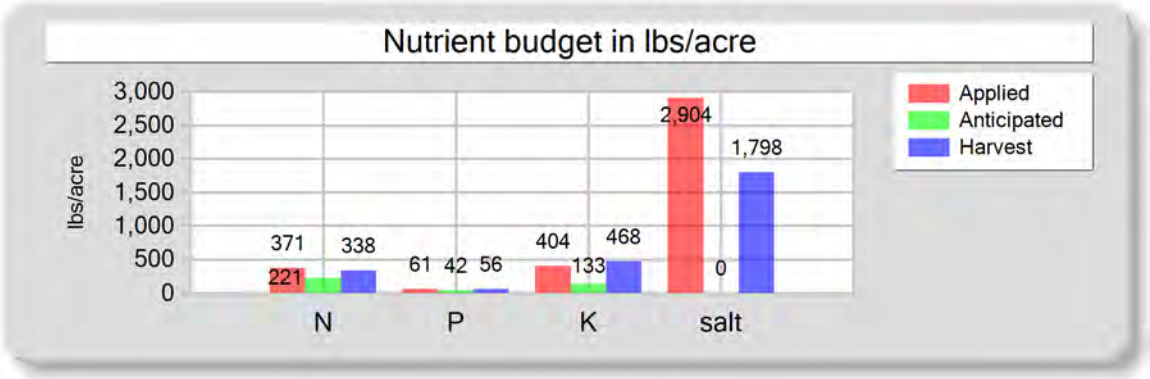
Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
3.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4 - 10/27/2022: Wheat, silage, soft dough

Field name: 4 Crop: Wheat, silage, soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	360.44	61.13	404.34	2,383.23
Fresh water	3.41	0.00	0.00	521.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	370.85	61.13	404.34	2,904.47
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	338.33	55.62	468.10	1,798.24
Nutrient balance	32.53	5.52	-63.76	1,106.23
Applied to removed ratio	1.10	1.10	0.86	1.62

Fresh water applied	
12,852,000.00	*gallons*
473.30	*acre-inches*
9.47	*inches/acre*

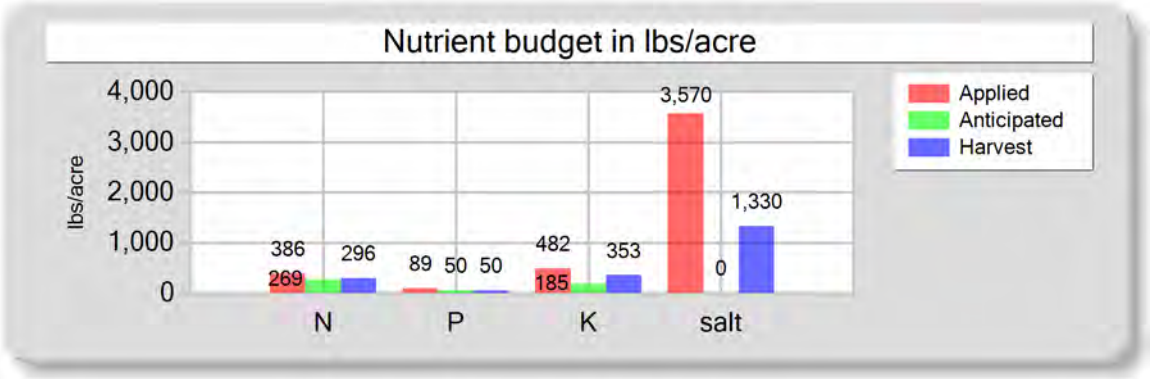
Process wastewater applied	
5,940,000.00	*gallons*
218.75	*acre-inches*
4.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4 - 05/18/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.41	50.07	187.03	0.00
Process wastewater	199.72	39.28	295.33	1,607.35
Fresh water	12.84	0.00	0.00	1,962.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	385.97	89.35	482.36	3,569.64
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	295.59	50.47	353.26	1,329.71
Nutrient balance	90.38	38.89	129.10	2,239.93
Applied to removed ratio	1.31	1.77	1.37	2.68

Fresh water applied	
48,384,000.00	*gallons*
1,781.82	*acre-inches*
35.64	*inches/acre*

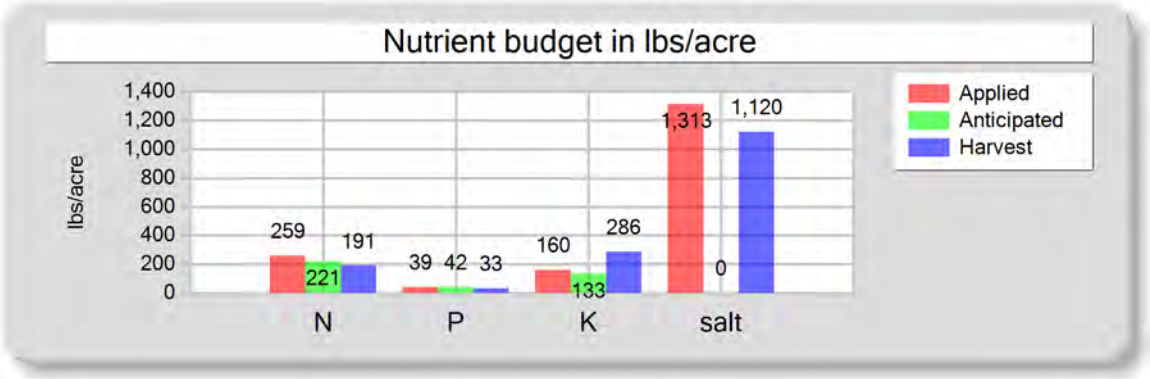
Process wastewater applied	
3,540,000.00	*gallons*
130.37	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 10/17/2022: Wheat, silage, soft dough

Field name: 5 Crop: Wheat, silage, soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	136.04	19.43	31.09	0.00
Process wastewater	114.95	20.02	129.12	761.87
Fresh water	3.61	0.00	0.00	551.23
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	259.26	39.46	160.21	1,313.10
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	190.67	33.00	286.01	1,120.39
Nutrient balance	68.59	6.46	-125.80	192.70
Applied to removed ratio	1.36	1.20	0.56	1.17

Fresh water applied	
32,076,000.00	*gallons*
1,181.25	*acre-inches*
10.01	*inches/acre*

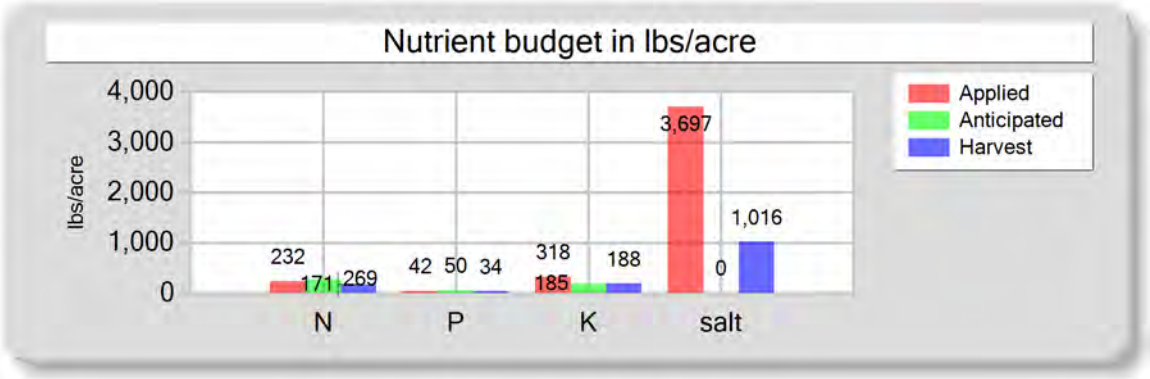
Process wastewater applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 04/26/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 04/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.99	42.21	317.60	1,722.75
Fresh water	12.92	0.00	0.00	1,973.84
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	231.58	42.21	317.60	3,696.59
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	171.12	34.22	188.23	1,015.87
Nutrient balance	60.46	7.99	129.37	2,680.72
Applied to removed ratio	1.35	1.23	1.69	3.64

Fresh water applied	
114,858,000.00	*gallons*
4,229.83	*acre-inches*
35.85	*inches/acre*

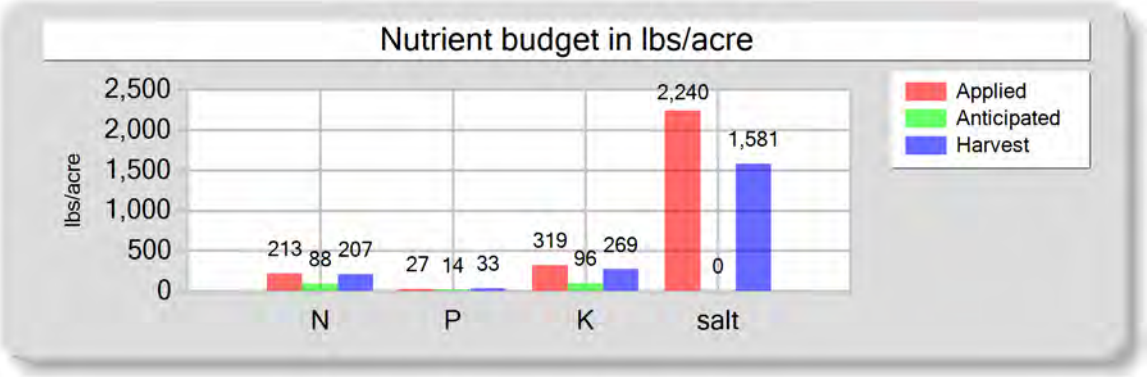
Process wastewater applied	
9,120,000.00	*gallons*
335.86	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 09/04/2023: Sudangrass, silage

Field name: 5 Crop: Sudangrass, silage Plant date: 09/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.40	27.35	318.70	1,736.16
Fresh water	3.30	0.00	0.00	503.90
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	213.37	27.35	318.70	2,240.06
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	207.12	33.05	268.81	1,580.93
Nutrient balance	6.25	-5.70	49.89	659.12
Applied to removed ratio	1.03	0.83	1.19	1.42

Fresh water applied	
29,322,000.00	*gallons*
1,079.83	*acre-inches*
9.15	*inches/acre*

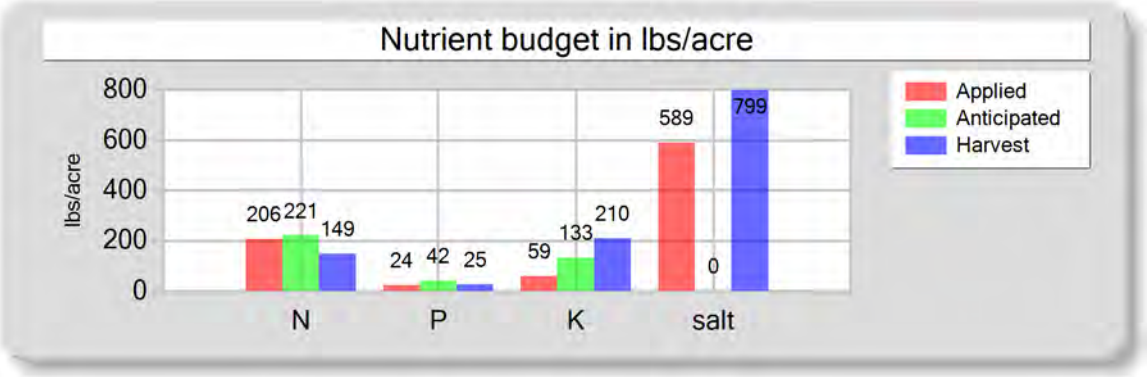
Process wastewater applied	
10,860,000.00	*gallons*
399.94	*acre-inches*
3.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 10/18/2022: Wheat, silage, boot stage

Field name: 6 Crop: Wheat, silage, boot stage Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.30	24.16	58.67	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.26	0.00	0.00	588.59
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	206.23	24.16	58.67	588.59
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	148.71	25.39	210.38	798.78
Nutrient balance	57.51	-1.23	-151.71	-210.19
Applied to removed ratio	1.39	0.95	0.28	0.74

Fresh water applied	
25,500,000.00	*gallons*
939.08	*acre-inches*
9.99	*inches/acre*

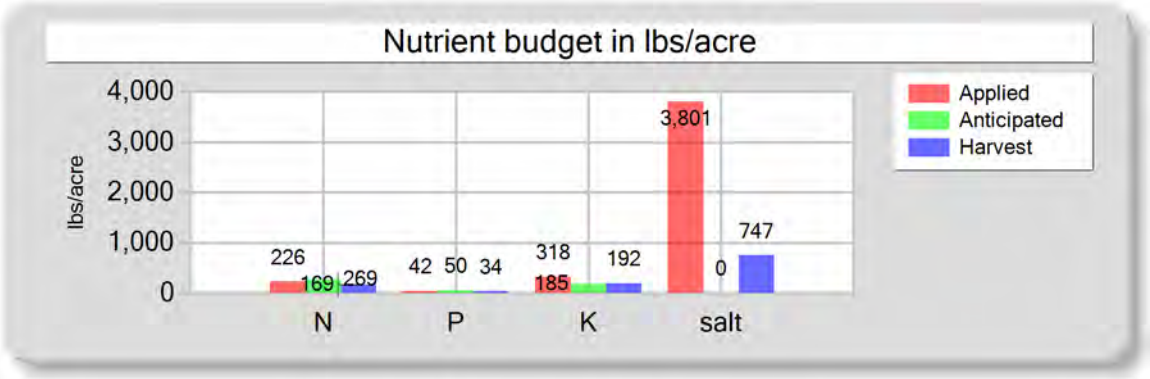
Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 04/28/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.56	42.24	318.07	1,719.74
Fresh water	8.00	0.00	0.00	2,081.25
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	226.23	42.24	318.07	3,800.99
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	169.27	33.85	191.84	747.49
Nutrient balance	56.96	8.39	126.23	3,053.50
Applied to removed ratio	1.34	1.25	1.66	5.08

Fresh water applied	
90,168,000.00	*gallons*
3,320.58	*acre-inches*
35.33	*inches/acre*

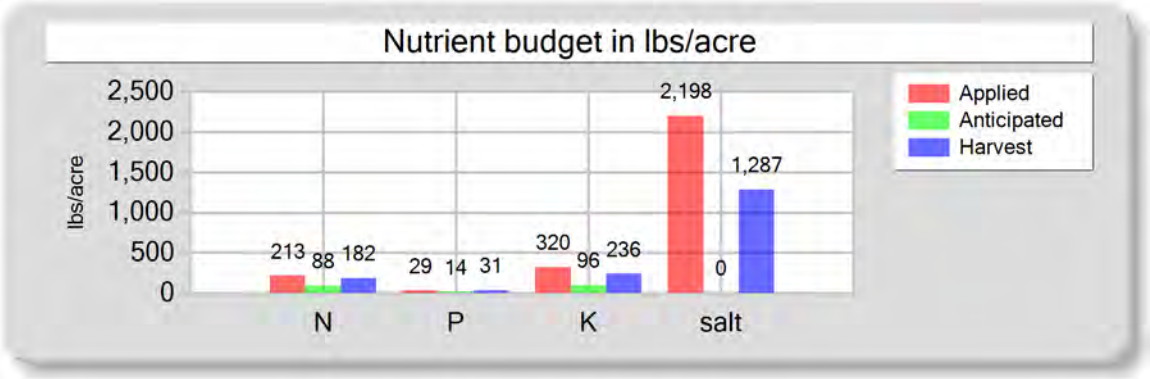
Process wastewater applied	
7,380,000.00	*gallons*
271.78	*acre-inches*
2.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 09/01/2023: Sudangrass, silage

Field name: 6 Crop: Sudangrass, silage Plant date: 09/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.47	28.70	320.44	1,738.97
Fresh water	1.77	0.00	0.00	459.10
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	212.90	28.70	320.44	2,198.07
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	181.91	30.64	235.53	1,286.81
Nutrient balance	30.99	-1.94	84.91	911.26
Applied to removed ratio	1.17	0.94	1.36	1.71

Fresh water applied	
19,890,000.00	*gallons*
732.48	*acre-inches*
7.79	*inches/acre*

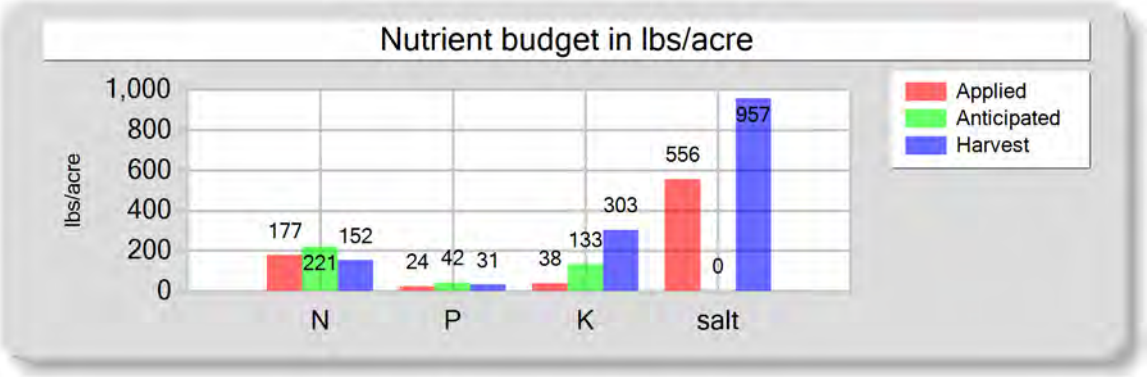
Process wastewater applied	
8,700,000.00	*gallons*
320.39	*acre-inches*
3.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7 - 10/21/2022: Wheat, silage, soft dough

Field name: 7 Crop: Wheat, silage, soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	167.90	23.99	38.38	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.25	0.00	0.00	555.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.14	23.99	38.38	555.64
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	151.66	31.06	303.32	957.48
Nutrient balance	25.48	-7.08	-264.95	-401.84
Applied to removed ratio	1.17	0.77	0.13	0.58

Fresh water applied	
18,870,000.00	*gallons*
694.92	*acre-inches*
9.93	*inches/acre*

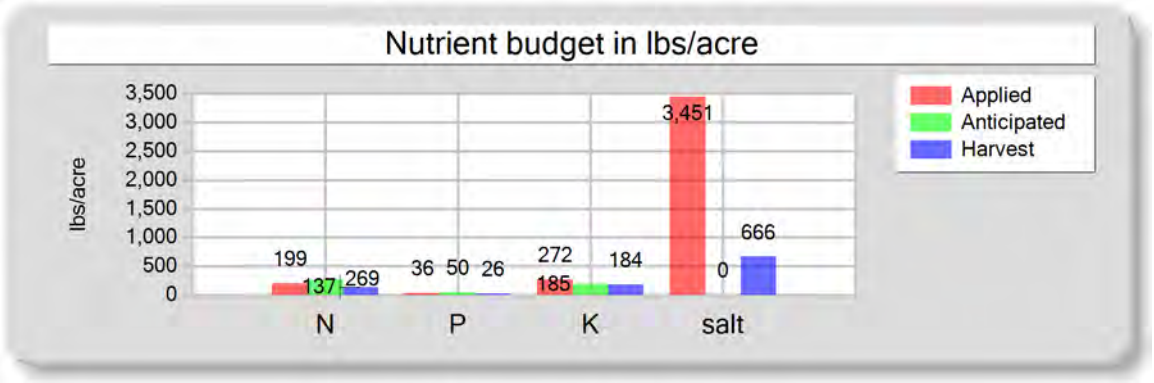
Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7 - 05/03/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.94	36.17	271.95	1,480.36
Fresh water	7.98	0.00	0.00	1,970.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.92	36.17	271.95	3,450.64
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	136.63	25.62	183.60	666.07
Nutrient balance	62.29	10.56	88.36	2,784.57
Applied to removed ratio	1.46	1.41	1.48	5.18

Fresh water applied	
66,912,000.00	*gallons*
2,464.14	*acre-inches*
35.20	*inches/acre*

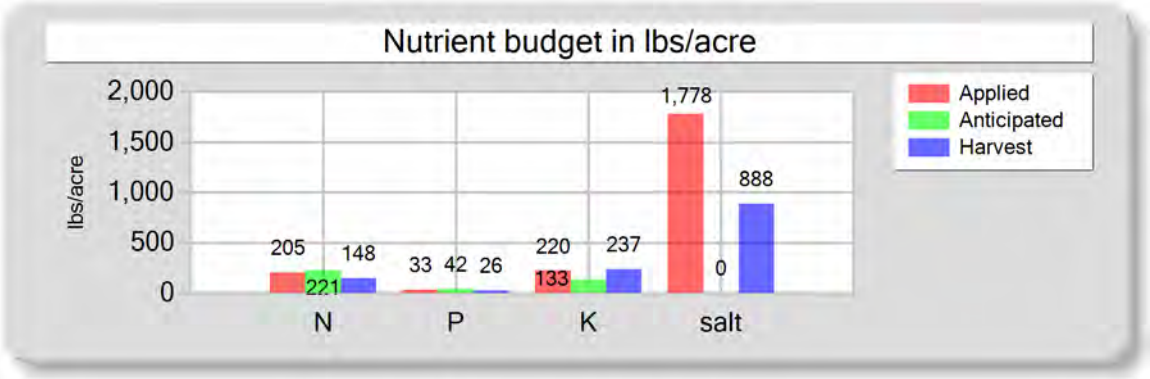
Process wastewater applied	
4,560,000.00	*gallons*
167.93	*acre-inches*
2.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8 - 10/24/2022: Wheat, silage, soft dough

Field name: 8 Crop: Wheat, silage, soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.43	33.22	220.32	1,298.45
Fresh water	1.94	0.00	0.00	479.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.37	33.22	220.32	1,777.56
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	147.86	25.88	236.58	887.77
Nutrient balance	57.51	7.35	-16.26	889.79
Applied to removed ratio	1.39	1.28	0.93	2.00

Fresh water applied	
24,174,000.00	*gallons*
890.25	*acre-inches*
8.56	*inches/acre*

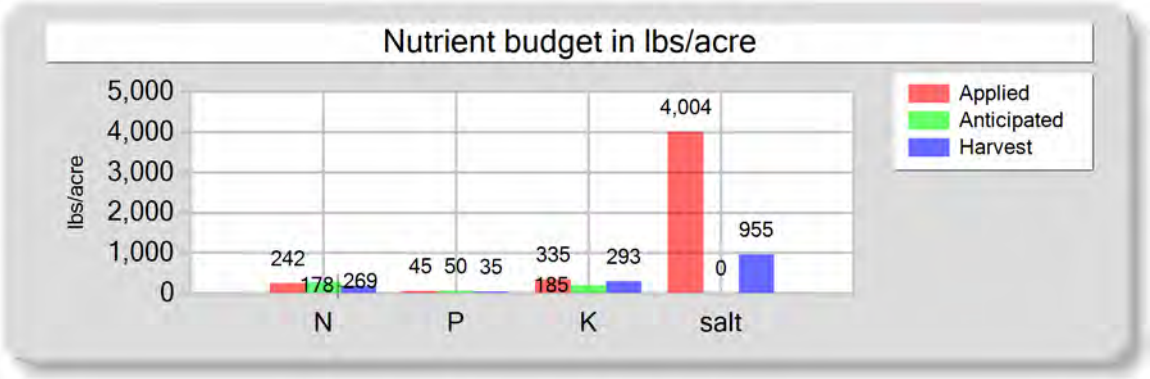
Process wastewater applied	
6,840,000.00	*gallons*
251.89	*acre-inches*
2.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8 - 05/05/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.23	44.54	335.00	1,820.91
Fresh water	8.84	0.00	0.00	2,183.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.07	44.54	335.00	4,004.21
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	178.26	34.50	293.26	955.17
Nutrient balance	63.81	10.04	41.74	3,049.04
Applied to removed ratio	1.36	1.29	1.14	4.19

Fresh water applied	
110,160,000.00	*gallons*
4,056.82	*acre-inches*
39.01	*inches/acre*

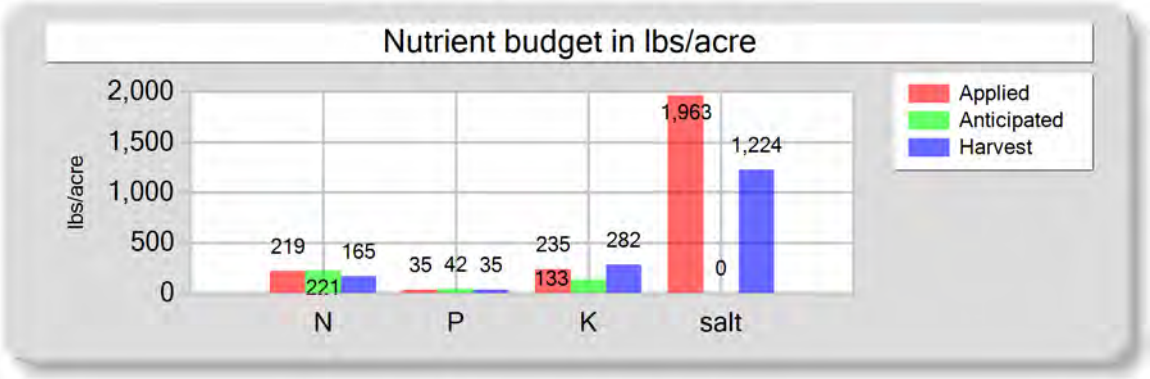
Process wastewater applied	
8,400,000.00	*gallons*
309.34	*acre-inches*
2.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

9 - 10/25/2022: Wheat, silage, soft dough

Field name: 9 Crop: Wheat, silage, soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.91	35.25	235.36	1,386.72
Fresh water	2.33	0.00	0.00	575.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.24	35.25	235.36	1,962.61
Anticipated crop nutrient removal	220.80	41.60	132.80	0.00
Actual crop nutrient removal	164.78	34.69	281.87	1,223.53
Nutrient balance	54.46	0.56	-46.51	739.08
Applied to removed ratio	1.33	1.02	0.84	1.60

Fresh water applied	
12,852,000.00	*gallons*
473.30	*acre-inches*
10.29	*inches/acre*

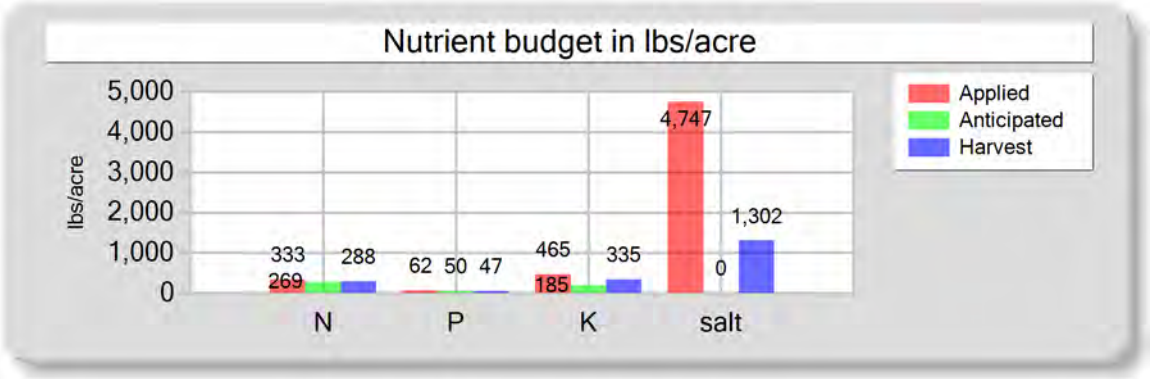
Process wastewater applied	
3,360,000.00	*gallons*
123.74	*acre-inches*
2.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

9 - 05/08/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	311.94	61.71	464.67	2,511.99
Fresh water	13.99	0.00	0.00	2,235.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	332.93	61.71	464.67	4,747.03
Anticipated crop nutrient removal	268.80	50.40	184.80	0.00
Actual crop nutrient removal	287.92	47.01	334.93	1,301.53
Nutrient balance	45.00	14.70	129.74	3,445.50
Applied to removed ratio	1.16	1.31	1.39	3.65

Fresh water applied	
50,082,000.00	*gallons*
1,844.35	*acre-inches*
40.09	*inches/acre*

Process wastewater applied	
5,280,000.00	*gallons*
194.44	*acre-inches*
4.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Corral Solids

Sample and source description: 2022 Fall Corral Solids
Sample date: 09/28/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 6.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	7,700.00	30,800.00							0.00
DL	100.00	100.00	100.00							0.01

2022 Fall Drying Solids

Sample and source description: 2022 Fall Drying Solids
Sample date: 09/28/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 18.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	2,800.00	6,800.00							0.00
DL	100.00	100.00	100.00							0.01

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids
Sample date: 09/28/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	2,500.00	4,000.00							0.00
DL	100.00	100.00	100.00							0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2023 Spring Corral Solids

Sample and source description: 2023 Spring Corral Solids
Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 19.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	8,200.00	27,400.00							0.00
DL	100.00	100.00	100.00							0.01

2023 Spring Drying Solids

Sample and source description: 2023 Spring Drying Solids
Sample date: 04/19/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 23.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	6,800.00	25,400.00							0.00
DL	100.00	100.00	100.00							0.01

2023 Spring Separator Solids

Sample and source description: 2023 Spring Separator Solids
Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 78.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	2,100.00	4,700.00							0.00
DL	100.00	100.00	100.00							0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2023 Fall Corral Solids

Sample and source description: 2023 Fall Corral Solids
Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 14.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	5,200.00	30,700.00	22,100.00	12,200.00	3,700.00	4,500.00	2,100.00		33.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

2023 Fall Drying Solids

Sample and source description: 2023 Fall Drying Solids
Sample date: 10/13/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 54.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,400.00	4,000.00	7,500.00	12,400.00	4,800.00	1,600.00	5,300.00	2,000.00		18.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

2023 Fall Separated Solids

Sample and source description: 2023 Fall Separated Solids
Sample date: 10/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 80.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	3,400.00	15,600.00	12,000.00	6,800.00	2,200.00	3,000.00	4,000.00		10.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2022 WWS Q4

Sample and source description: 2022 WWS Q4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	293.00	218.00	0.00	1.80	47.60	328.00								4,300.00	1,930
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

2023 WWS Q1

Sample and source description: 2023 WWS Q1

Sample date: 02/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	430.00	292.00	0.00	2.20	74.90	483.00								6,060.00	2,850
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

2023 WWS Q2

Sample and source description: 2023 WWS Q2

Sample date: 06/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	391.00	352.00	0.00	21.90	75.30	563.00								4,940.00	3,140
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

2023 WWS Q3

Sample and source description: 2023 WWS Q3

Sample date: 09/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	151.00	136.00	0.00	0.20	20.20	233.00	56.00	48.80	132.00	1,190.00	0.00	18.00	86.80	2,540.00	1,580
DL	1.00	0.50	0.50	0.10	0.10	0.50	0.10	0.10	1.00	10.00	1.00	0.50	0.20	10.00	10

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

2023 WWS Q3

Sample and source description: 2023 WWS Q3

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	266.00	236.00	0.00	0.30	54.50	414.00								4,600.00	2,150
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

2023 WWS Q4

Sample and source description: 2023 WWS Q4

Sample date: 10/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	269.00	257.00	0.00	0.90	15.20	416.00								5,060.00	2,380
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Ag Supply Well

IW-A

Sample description: IW-A

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.32	0.00	0.80								420.00	250
DL	1.00	0.50	0.10								10.00	10

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Ag Supply Well

IW-B

Sample description: IW-B

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.80								414.00	247
DL	1.00	0.50	0.10								10.00	10

IW-C

Sample description: IW-C

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.90								436.00	260
DL	1.00	0.50	0.10								10.00	10

IW-D

Sample description: IW-D

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.54	0.00	0.80								424.00	246
DL	1.00	0.50	0.10								10.00	10

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Ag Supply Well

IW-L

Sample description: IW-L

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.59	0.00	0.90								431.00	243
DL	1.00	0.50	0.10								10.00	10

MID Canal

MID Canal

Sample description: MID Canal

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.10								36.00	27
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

1

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			221.00				
DL			2.00				

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,185.00				
DL			2.00				

11

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			651.00				
DL			2.00				

12

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			874.00				
DL			2.00				

13

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			557.00				
DL			2.00				

2

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/23/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			179.00				
DL			2.00				

3

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/23/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			426.00				
DL			2.00				

4

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			251.00				
DL			2.00				

5

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			275.00				
DL			2.00				

6

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			253.00				
DL			2.00				

7

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

7

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,084.00				
DL			2.00				

8

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			252.00				
DL			2.00				

9

Soil Sample

Sample and source description: Soil Sample

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			786.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

1 - 11/07/2022: Wheat, silage, soft dough

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

1 - 11/07/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	1,200.00	11,000.00		10.30
DL	100.00	100.00	100.00		0.01

1 - 05/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 76.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,900.00	500.00	3,200.00		6.20
DL	100.00	100.00	100.00		0.01

10 - 11/01/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	8,000.00		11.60
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

10 - 05/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	5,900.00		4.30
DL	100.00	100.00	100.00		0.01

11 - 11/03/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	800.00		11.60
DL	100.00	100.00	100.00		0.01

11 - 05/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	700.00	5,800.00		4.60
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

12 - 11/02/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	800.00		11.60
DL	100.00	100.00	100.00		0.01

12 - 05/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	3,800.00		5.50
DL	100.00	100.00	100.00		0.01

13 - 11/04/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	800.00		11.60
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

13 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	1,000.00	5,300.00		6.50
DL	100.00	100.00	100.00		0.01

2 - 11/09/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	1,200.00	10,100.00		11.40
DL	100.00	100.00	100.00		0.01

2 - 05/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 76.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	700.00	3,300.00		6.70
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

3 - 11/08/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,700.00	1,300.00	12,600.00		10.50
DL	100.00	100.00	100.00		0.01

3 - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	700.00	4,000.00		6.40
DL	100.00	100.00	100.00		0.01

4 - 10/27/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	1,200.00	10,100.00		9.70
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

4 - 05/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	700.00	4,900.00		5.30
DL	100.00	100.00	100.00		0.01

5 - 10/17/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	900.00	7,800.00		10.50
DL	100.00	100.00	100.00		0.01

5 - 04/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	600.00	3,300.00		6.50
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

5 - 09/04/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan
Sample date: 12/06/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	1,500.00	12,200.00		20.50
DL	100.00	100.00	100.00		0.01

6 - 10/18/2022: Wheat, silage, boot stage

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	700.00	5,800.00		9.10
DL	100.00	100.00	100.00		0.01

6 - 04/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	600.00	3,400.00		4.80
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

6 - 09/01/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan
Sample date: 12/06/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	1,600.00	12,300.00		19.20
DL	100.00	100.00	100.00		0.01

7 - 10/21/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,300.00	1,700.00	16,600.00		13.10
DL	100.00	100.00	100.00		0.01

7 - 05/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	600.00	4,300.00		5.00
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

8 - 10/24/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	700.00	6,400.00		7.90
DL	100.00	100.00	100.00		0.01

8 - 05/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	600.00	5,100.00		4.90
DL	100.00	100.00	100.00		0.01

9 - 10/25/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	800.00	6,500.00		9.50
DL	100.00	100.00	100.00		0.01

9 - 05/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	800.00	5,700.00		5.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

No subsurface (tile) drainage analyses entered.

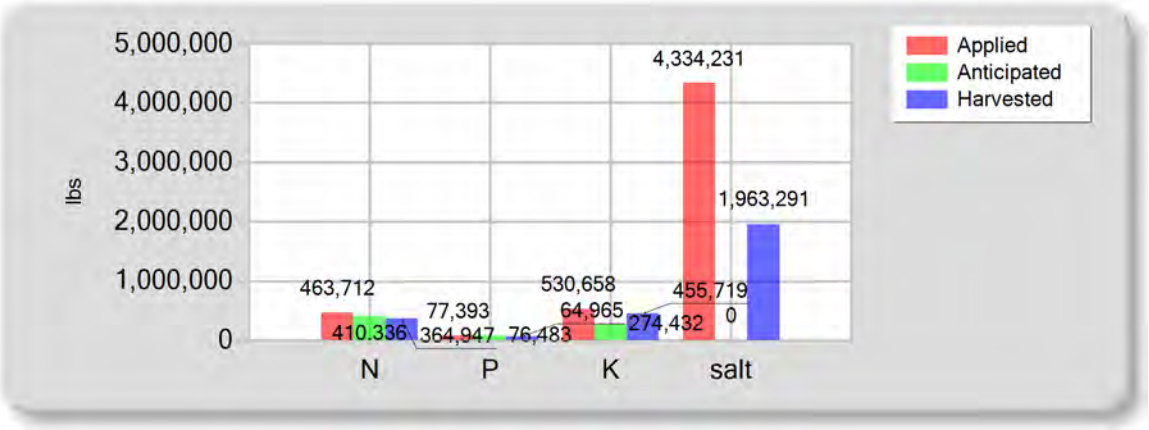
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

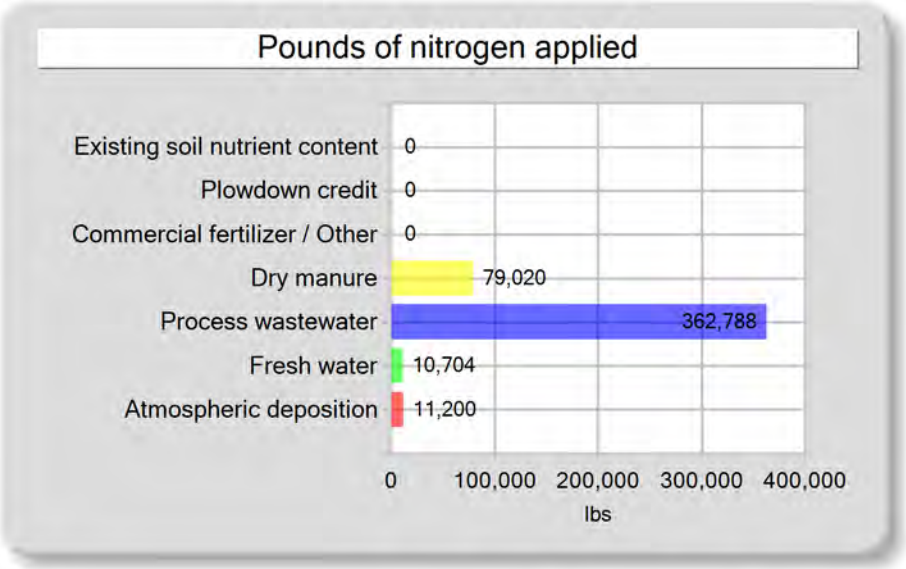
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79,019.83	12,430.58	40,599.19	0.00
Process wastewater	362,788.19	64,962.67	490,058.62	2,735,183.90
Fresh water	10,703.99	0.00	0.00	1,599,047.55
Atmospheric deposition	11,200.00	0.00	0.00	0.00
Total nutrients applied	463,712.02	77,393.25	530,657.81	4,334,231.45
Anticipated crop nutrient removal	410,336.00	76,483.20	274,432.00	0.00
Actual crop nutrient removal	364,947.00	64,965.24	455,719.46	1,963,290.71
Nutrient balance	98,765.02	12,428.01	74,938.35	2,370,940.75
Applied to removed ratio	1.27	1.19	1.16	2.21

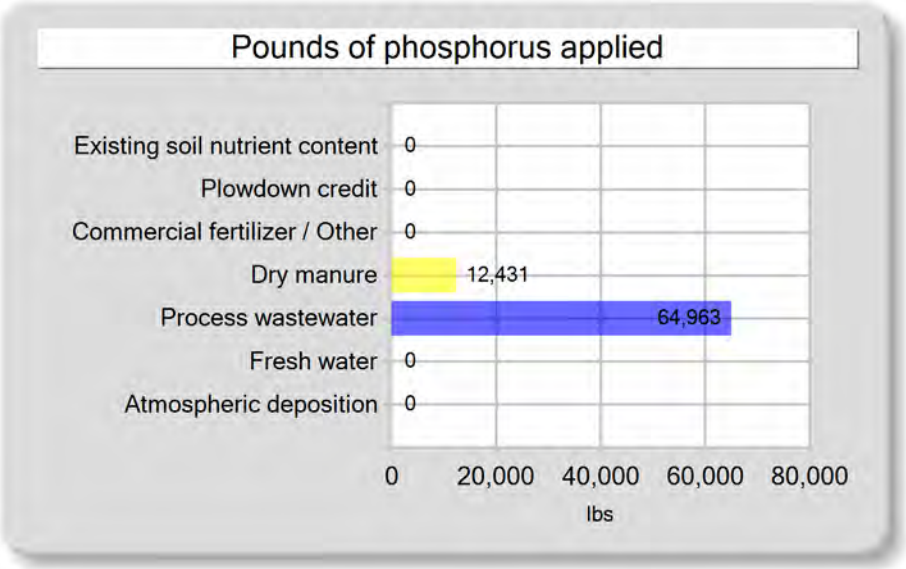
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

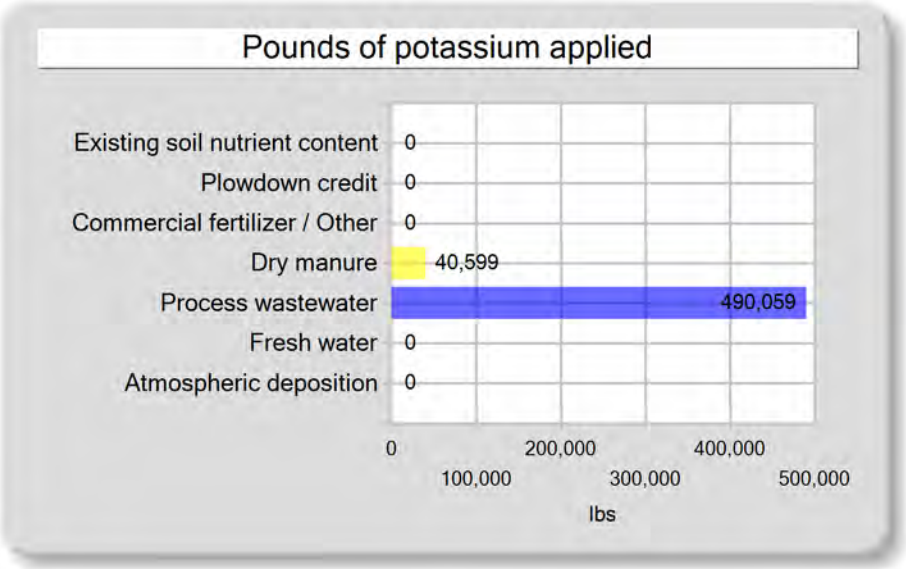


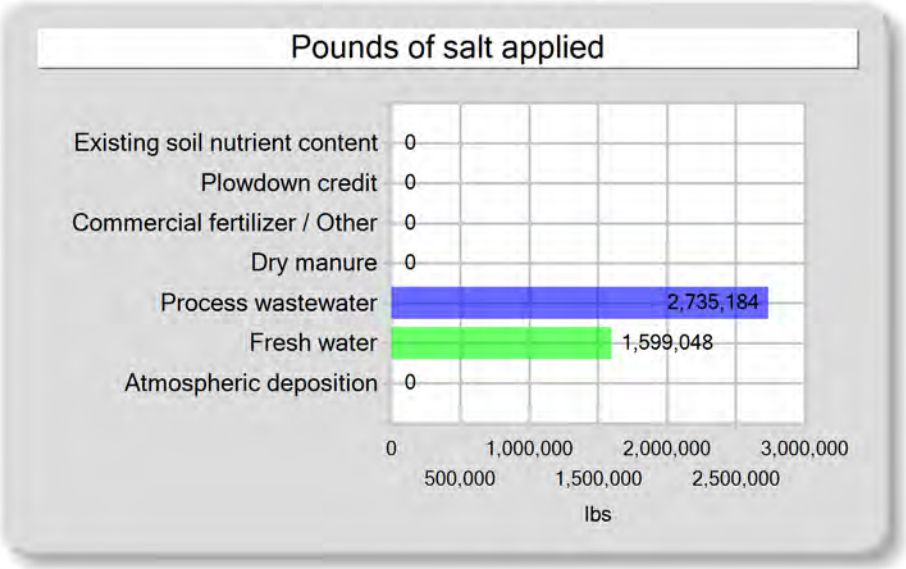
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ADDITIONAL NOTES

A. NOTES

Precipitation utilized during winter months to meet forage freshwater requirements. Heavy rains during the winter season allowed for a greater amount of surface water allocation to grow crops.

Fields 1 Wheat, 3 Wheat, 4 Wheat, 5 Sudan, & 6 Sudan had higher than anticipated removal rates. This was due to higher than expected %N and/or yield. The %N removed, which is based on analysis derived through a certified laboratory, exceeded the anticipated values for this crop. Appropriate management practices resulted in decent yields and/or higher than expected nutrient removal rates.

Field 6 Wheat, 7 Wheat, 8 Wheat, 12 Wheat, 1 Corn, 2 Corn, 5 Corn, 6 Corn, 8 Corn, & 12 Corn had lower than anticipated removal rates. This was due to a lower than expected %N &/or tonnage. The %N was based on analysis that was derived through a certified laboratory. However, the applications to these fields matched the low removal rates and was able to meet the field ratio threshold of 1.4.

Field 7 Corn had a lower than anticipated removal rate, which was due to lower than expected %N and tonnage. The %N was based on analysis that was derived through a certified laboratory. The low removal rates resulted in a field ratio slightly exceeding target limits, even though application rates did not exceed expected removals.

Majority of the solid manure generated was windrowed and used for bedding for newly built barns.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Larry Meirinho	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/26/2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X016-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 10/13/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,785.00 tons

Manure Solids Content: 45.8 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...
6/26/2024

Operator Signature — Date

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...
6/26/2024

Hauler Signature — Date

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X005-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 01/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 8,300,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...

6/26/2024

Operator Signature — Date

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...

6/26/2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

Number and Street	City	County	Zip Code
4890 Healy RD	Merced	Merced	95341

Contact Person Name and Phone Number: Larry Meirinho (Name) (209) 652-9780 (Phone Number)

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
4890 Healy RD	Merced	CA	95341

Contact Person: Larry Meirinho (Name) (209) 652-9780 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name: Lazy J Farms
Phone Number: (209) 652-9780

Address	City	State	Zip Code
4890 Healy RD	Merced	CA	95341

Destination Address or Assessor's Parcel Number:

Address:
City:
Zip Code:

Street and nearest cross street (if no address):
County:

Assessor's Parcel Number: X066-X170-X005-XXXX
Assessor's Parcel Number County: Merced

Last date hauled: 03/07/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 9,500,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
6/26/2024

Operator Signature — Date

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
6/26/2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X005-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 06/07/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 7,650,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...

6/26/2024

Operator Signature — Date

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...

6/26/2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X016-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 06/13/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 6,500,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...
6/26/2024

Operator Signature — Date

DocuSigned by:
Larry Meirinks
AEDA3F36DB6A4E7...
6/26/2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X005-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 07/23/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 6,750,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
Operator Signature — Date: 6/26/2024

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
Hauler Signature — Date: 6/26/2024

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Larry Meirinho

Name of Dairy Facility: Meirinho Holsteins Inc

Facility Address:

4890 Healy RD	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Larry Meirinho (209) 652-9780
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Larry Meirinho

Address of Hauling Company/Person:

4890 Healy RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Larry Meirinho (209) 652-9780
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lazy J Farms (209) 652-9780
Name Phone Number

4890 Healy RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X066-X170-X016-XXXX Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 08/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 5,400,000 gallons

Method used to determine volume of process wastewater:

Pump rate (GPM) multiplied by run time

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[ ] YES [X] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
Operator Signature — Date: 6/26/2024

DocuSigned by:
Larry Meirinho
AEDA3F36DB6A4E7...
Hauler Signature — Date: 6/26/2024

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™
LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0150-01	IW A	Ag Water	April	Fresh Water	11/02/2023 8:00
23K0150-02	IW B	Ag Water	April	Fresh Water	11/02/2023 8:05
23K0150-03	IW C	Ag Water	April	Fresh Water	11/02/2023 8:10
23K0150-04	IW D	Ag Water	April	Fresh Water	11/02/2023 8:15
23K0150-05	IW L	Ag Water	April	Fresh Water	11/02/2023 8:20
23K0150-06	MID Canal	Ag Water	April	Fresh Water	11/02/2023 8:30

Default Cooler Temperature on Receipt °C: 2.6

Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results

Sample: IW A
23K0150-01 (Water)

Sampled: 11/2/2023 8:00
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.42**	mmhos/cm	0.01	1		11/06/23 10:15	SM 2510 B		BEK0126
Electrical Conductivity umhos	**420**	umhos/cm	10.0	1		11/06/23 10:15	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	**0.8**	mg/L	0.1	1	10	11/03/23 19:50	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:15	SM 4500-H+	H	BEK0126
pH	**7.9**	units	1.0	1		11/06/23 10:15	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**250**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:30	SM 4500-NH3 C		BEK0101
Total Nitrogen	**1.32**	mg/L	1.00	1		11/08/23 08:30	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results
(Continued)

Sample: IW B
23K0150-02 (Water)

Sampled: 11/2/2023 8:05
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.41**	mmhos/cm	0.01	1		11/06/23 10:17	SM 2510 B		BEK0126
Electrical Conductivity umhos	**414**	umhos/cm	10.0	1		11/06/23 10:17	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	**0.8**	mg/L	0.1	1	10	11/03/23 20:10	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:17	SM 4500-H+	H	BEK0126
pH	**7.9**	units	1.0	1		11/06/23 10:17	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**247**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:32	SM 4500-NH3 C		BEK0101
Total Nitrogen	ND	mg/L	1.00	1		11/08/23 08:32	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results
(Continued)

Sample: IW C
23K0150-03 (Water)

Sampled: 11/2/2023 8:10
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.44**	mmhos/cm	0.01	1		11/06/23 10:23	SM 2510 B		BEK0126
Electrical Conductivity umhos	**436**	umhos/cm	10.0	1		11/06/23 10:23	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	**0.9**	mg/L	0.1	1	10	11/03/23 20:31	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:23	SM 4500-H+	H	BEK0126
pH	**7.9**	units	1.0	1		11/06/23 10:23	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**260**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:33	SM 4500-NH3 C		BEK0101
Total Nitrogen	ND	mg/L	1.00	1		11/08/23 08:33	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results
(Continued)

Sample: IW D
23K0150-04 (Water)

Sampled: 11/2/2023 8:15
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.42**	mmhos/cm	0.01	1		11/06/23 10:25	SM 2510 B		BEK0126
Electrical Conductivity umhos	**424**	umhos/cm	10.0	1		11/06/23 10:25	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	**0.8**	mg/L	0.1	1	10	11/03/23 20:51	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:25	SM 4500-H+	H	BEK0126
pH	**7.9**	units	1.0	1		11/06/23 10:25	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**246**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:36	SM 4500-NH3 C		BEK0101
Total Nitrogen	**1.54**	mg/L	1.00	1		11/08/23 08:36	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results
(Continued)

Sample: IW L
23K0150-05 (Water)

Sampled: 11/2/2023 8:20
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.43**	mmhos/cm	0.01	1		11/06/23 10:26	SM 2510 B		BEK0126
Electrical Conductivity umhos	**431**	umhos/cm	10.0	1		11/06/23 10:26	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	**0.9**	mg/L	0.1	1	10	11/03/23 21:11	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:26	SM 4500-H+	H	BEK0126
pH	**7.9**	units	1.0	1		11/06/23 10:26	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**243**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:37	SM 4500-NH3 C		BEK0101
Total Nitrogen	**1.59**	mg/L	1.00	1		11/08/23 08:37	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Sample Results
(Continued)

Sample: MID Canal
23K0150-06 (Water)

Sampled: 11/2/2023 8:30
Sampled By: April

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.04**	mmhos/cm	0.01	1		11/06/23 10:28	SM 2510 B		BEK0126
Electrical Conductivity umhos	**36.0**	umhos/cm	10.0	1		11/06/23 10:28	SM 2510 B		BEK0126
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/03/23 21:31	EPA 300.0		BEK0071
Temperature	**25.0**	units	0.0	1		11/06/23 10:28	SM 4500-H+	H	BEK0126
pH	**7.3**	units	1.0	1		11/06/23 10:28	SM 4500-H+	H	BEK0126
Total Filterable Solids (TDS)	**27.0**	mg/L	10.0	1		11/06/23 14:33	SM 2540 C		BEK0103
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/08/23 08:39	SM 4500-NH3 C		BEK0101
Total Nitrogen	ND	mg/L	1.00	1		11/08/23 08:39	SM 4500-NH3 C		BEK0101

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0071										
Blank (BEK0071-BLK1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0071-BLK2)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0071-BLK3)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0071-BS1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEK0071-BS2)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
Duplicate (BEK0071-DUP1)		**Source: 23K0150-01**			Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	0.8		0.1	mg/L		0.8			1.07	10
Duplicate (BEK0071-DUP2)		**Source: 23K0153-03**			Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	8.2		0.1	mg/L		8.2			0.451	10
Matrix Spike (BEK0071-MS1)		**Source: 23K0150-01**			Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	6.0		0.1	mg/L	5.000	0.8	103	90-110		
Matrix Spike (BEK0071-MS2)		**Source: 23K0153-03**			Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	13.3		0.1	mg/L	5.000	8.2	103	90-110		
Reference (BEK0071-SRM1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0071-SRM2)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		
Reference (BEK0071-SRM3)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0101										
Blank (BEK0101-BLK1)					Prepared: 11/3/2023	Analyzed: 11/7/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEK0101-BLK2)					Prepared: 11/3/2023	Analyzed: 11/7/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEK0101-BS1)					Prepared: 11/3/2023	Analyzed: 11/7/2023				
Kjeldahl Nitrogen (TKN), Total	5.34		1.00	mg/L	5.709		93.5	90-110		
LCS (BEK0101-BS2)					Prepared: 11/3/2023	Analyzed: 11/8/2023				
Kjeldahl Nitrogen (TKN), Total	5.23		1.00	mg/L	5.709		91.6	90-110		
Duplicate (BEK0101-DUP1)		**Source: 23K0002-01**			Prepared: 11/3/2023	Analyzed: 11/7/2023				
Kjeldahl Nitrogen (TKN), Total	157		3.50	mg/L		153			2.81	10
Duplicate (BEK0101-DUP2)		**Source: 23K0161-02**			Prepared: 11/3/2023	Analyzed: 11/8/2023				
Kjeldahl Nitrogen (TKN), Total	198		3.50	mg/L		197			0.726	10
Matrix Spike (BEK0101-MS1)		**Source: 23K0002-01**			Prepared: 11/3/2023	Analyzed: 11/7/2023				
Kjeldahl Nitrogen (TKN), Total	213		3.50	mg/L	9.990	153	602	90-110		
Matrix Spike (BEK0101-MS2)		**Source: 23K0161-02**			Prepared: 11/3/2023	Analyzed: 11/8/2023				
Kjeldahl Nitrogen (TKN), Total	196		3.50	mg/L	9.990	197	NR	90-110		
Reference (BEK0101-SRM1)					Prepared: 11/3/2023	Analyzed: 11/8/2023				
Kjeldahl Nitrogen (TKN), Total	20.9			mg/L	23.80		87.6	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0103										
Blank (BEK0103-BLK1)					Prepared: 11/3/2023 Analyzed: 11/6/2023					
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEK0103-BS1)					Prepared: 11/3/2023 Analyzed: 11/6/2023					
Total Filterable Solids (TDS)	53.8		10.0	mg/L	2000		2.69	0-200		
Duplicate (BEK0103-DUP1)			**Source: 23K0150-01**		Prepared: 11/3/2023 Analyzed: 11/6/2023					
Total Filterable Solids (TDS)	250		10.0	mg/L		250			0.00	10
Duplicate (BEK0103-DUP2)			**Source: 23K0153-02**		Prepared: 11/3/2023 Analyzed: 11/6/2023					
Total Filterable Solids (TDS)	315		10.0	mg/L		320			1.57	10
Reference (BEK0103-SRM1)					Prepared: 11/3/2023 Analyzed: 11/6/2023					
Total Filterable Solids (TDS)	330			mg/L	325.0		102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0126										
Blank (BEK0126-BLK1)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.4		1.0	units						
Blank (BEK0126-BLK2)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0126-BLK3)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0126-DUP1)		**Source: 23K0150-01**			Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	0.41		0.01	mmhos/cm		0.42			1.20	10
pH	7.9		1.0	units		7.9			0.00	10
Electrical Conductivity umhos	414		10.0	umhos/cm		420			1.20	10
Duplicate (BEK0126-DUP2)		**Source: 23K0159-01**			Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	0.76		0.01	mmhos/cm		0.76			0.0394	10
Electrical Conductivity umhos	761		10.0	umhos/cm		761			0.0394	10
pH	7.7		1.0	units		7.7			0.130	10
Reference (BEK0126-SRM1)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	517			umhos/cm	538.0		96.1	90-110		
Reference (BEK0126-SRM2)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
pH	7.5			units	7.520		99.6	67021-101.32		
Reference (BEK0126-SRM3)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	961			umhos/cm	1000		96.1	90-110		
Electrical Conductivity umhos	961			umhos/cm	1000		96.1	90-110		
Reference (BEK0126-SRM4)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	960			umhos/cm	1000		96.0	90-110		
Electrical Conductivity umhos	960			umhos/cm	1000		96.0	90-110		
Reference (BEK0126-SRM5)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity	965			umhos/cm	1000		96.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/14/2023 10:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0126 (Continued)										
Reference (BEK0126-SRM5)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
Electrical Conductivity umhos	965			umhos/cm	1000		96.5	90-110		
Reference (BEK0126-SRM6)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0126-SRM7)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0126-SRM8)					Prepared: 11/3/2023	Analyzed: 11/6/2023				
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

11/02/23 16:00 23K0150

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 6 No of Bottles: 12

Purchase Order No

Bill To: 22203 Acct # | 8 Cons # 16

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Meirinho Holsteins LP
Address: 5418 Woodland Ave
City: Modesto **State:** CA **Zip:** 95358
Telephone: (209) 652-9780 **Fax:**
Cell/Email: akm5604@gmail.com
COPY TO: krhodes@fragservices.com
ariordan@fragservices.com
REQUESTED BY:
PROJECT:
CROP: Fresh water

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

( ) DWW1: EC, NO_3-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(✓) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

[X] Copy of Chain [X] QA/QC Documents

Sampled By: ~~FR Ag Services~~ April

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	IW A	11/2	0800	2.6	
2	IW B		0805	4.3	
3	IW C		0810	1.4	
4	IW D		0815	2.0	
5	IW L		0820	1.8	
6	MID Canal		0830	3.1	
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Kierstan Rhodes	FR Ag Services	11/2 0830	
Second	JCF	DLV	11-2-23 2:00	11-2-23 3:45
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Mileage
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[Meirinho Holsteins LP Water.xls]Dairy Water rev 3.6.18

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

Fast

11/02/23 16:00 23K0150

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic	1	1	1	1	1	1				
Plastics	* pH Value	<2	<2	<2	<2	<2	<2				
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic	1	1	1	1	1	1				
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

SEND OUT

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™
LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0140-01	Dom F	Ground Water	FR Ag Services	Fresh Water	11/02/2023 8:35
23K0140-02	Dom G	Ground Water	FR Ag Services	Fresh Water	11/02/2023 8:40
23K0140-03	Dom H	Ground Water	FR Ag Services	Fresh Water	11/02/2023 8:45
23K0140-04	Dom I	Ground Water	FR Ag Services	Fresh Water	11/02/2023 8:50
23K0140-05	Dom J	Ground Water	FR Ag Services	Fresh Water	11/02/2023 8:55
23K0140-06	Dom K	Ground Water	FR Ag Services	Fresh Water	11/02/2023 9:00

Default Cooler Temperature on Receipt °C: 2.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results

Sample: Dom F
23K0140-01 (Water)

Sampled: 11/2/2023 8:35
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.40**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**404**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	**0.9**	mg/L	0.1	1	10	11/03/23 19:02	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 11:59	SM 4500-H+	H	BEK0123
pH	**8.0**	units	1.0	1		11/14/23 11:59	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results
(Continued)

Sample: Dom G
23K0140-02 (Water)

Sampled: 11/2/2023 8:40
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.30**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**299**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/03/23 19:22	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 12:00	SM 4500-H+	H	BEK0123
pH	**8.2**	units	1.0	1		11/14/23 12:00	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results
(Continued)

Sample: Dom H
23K0140-03 (Water)

Sampled: 11/2/2023 8:45
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.66**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**664**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	**1.8**	mg/L	0.1	1	10	11/03/23 19:42	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 12:09	SM 4500-H+	H	BEK0123
pH	**8.1**	units	1.0	1		11/14/23 12:09	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results
(Continued)

Sample: Dom I
23K0140-04 (Water)

Sampled: 11/2/2023 8:50
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.30**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**299**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/03/23 20:01	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 12:10	SM 4500-H+	H	BEK0123
pH	**8.2**	units	1.0	1		11/14/23 12:10	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results
(Continued)

Sample: Dom J
23K0140-05 (Water)

Sampled: 11/2/2023 8:55
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.38**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**377**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	**0.7**	mg/L	0.1	1	10	11/03/23 20:21	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 12:12	SM 4500-H+	H	BEK0123
pH	**8.1**	units	1.0	1		11/14/23 12:12	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Sample Results
(Continued)

Sample: Dom K
23K0140-06 (Water)

Sampled: 11/2/2023 9:00
Sampled By: FR Ag Services

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.41**	mmhos/cm	0.01	1		11/14/23 14:22	SM 2510 B		BEK0443
Electrical Conductivity umhos	**409**	umhos/cm	10.0	1		11/14/23 14:22	SM 2510 B		BEK0443
Ammonia (as N)	*	mg/L	0.00	1		11/14/23 11:07	Field		BEK0429
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/03/23 20:41	EPA 300.0		BEK0070
Temperature	**25.0**	units	0.0	1		11/14/23 12:13	SM 4500-H+	H	BEK0123
pH	**7.3**	units	1.0	1		11/14/23 12:13	SM 4500-H+	H	BEK0123

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0070										
Blank (BEK0070-BLK1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0070-BLK2)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0070-BLK3)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0070-BS1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.8	90-110		
LCS (BEK0070-BS2)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
Duplicate (BEK0070-DUP1)		**Source: 23K0140-02**			Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L		ND				10
Duplicate (BEK0070-DUP2)		**Source: 23K0143-03**			Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	0.1		0.1	mg/L		0.1			1.92	10
Matrix Spike (BEK0070-MS1)		**Source: 23K0140-02**			Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	ND	94.9	90-110		
Matrix Spike (BEK0070-MS2)		**Source: 23K0143-03**			Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000	0.1	97.8	90-110		
Reference (BEK0070-SRM1)					Prepared & Analyzed: 11/3/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.8	90-110		
Reference (BEK0070-SRM2)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		100	90-110		
Reference (BEK0070-SRM3)					Prepared & Analyzed: 11/4/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0123										
Blank (BEK0123-BLK1)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
Temperature	25.0		0.0	units						
pH	5.8		1.0	units						
Blank (BEK0123-BLK2)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
Temperature	25.0		0.0	units						
pH	7.4		1.0	units						
Blank (BEK0123-BLK3)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Duplicate (BEK0123-DUP1)		**Source: 23K0140-02**			Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	8.2		1.0	units		8.2			0.122	10
Duplicate (BEK0123-DUP2)		**Source: 23K0142-05**			Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	8.0		1.0	units		8.0			0.125	10
Reference (BEK0123-SRM2)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	7.5			units	7.520		100	67021-101.3		
Reference (BEK0123-SRM6)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0123-SRM7)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0123-SRM8)					Prepared: 11/3/2023	Analyzed: 11/14/2023				
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Meirinho Holsteins LP
5418 Woodland Ave
Modesto, CA 95358

Account# 00-0022263
Account Manager: Ben Nydam
Submitted By:

Received: 11/02/2023 16:00
Reported: 11/16/2023 09:43

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0443										
Blank (BEK0443-BLK1)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0443-BLK2)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0443-BLK3)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0443-DUP1)		**Source: 23K0094-01**			Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	0.30		0.01	mmhos/cm		0.30			0.828	10
Electrical Conductivity umhos	301		10.0	umhos/cm		303			0.828	10
Duplicate (BEK0443-DUP2)		**Source: 23K0142-01**			Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	0.60		0.01	mmhos/cm		0.61			0.563	10
Electrical Conductivity umhos	602		10.0	umhos/cm		606			0.563	10
Reference (BEK0443-SRM1)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	538			umhos/cm	538.0		100	90-110		
Reference (BEK0443-SRM2)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	1000			umhos/cm	1000		100	90-110		
Electrical Conductivity umhos	1000			umhos/cm	1000		100	90-110		
Reference (BEK0443-SRM3)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	993			umhos/cm	1000		99.3	90-110		
Electrical Conductivity umhos	993			umhos/cm	1000		99.3	90-110		
Reference (BEK0443-SRM4)					Prepared & Analyzed: 11/14/2023					
Electrical Conductivity	1000			umhos/cm	1000		100	90-110		
Electrical Conductivity umhos	1000			umhos/cm	1000		100	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

11/02/23 16:00 23K0140

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 Fax 559 268-8174

Purchase Order No | Bill To: Acct # 22203 | Cons # 8

No. Samples: 6 No of Bottles: 12

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [X] Groundwater [ ] Monitoring Well

Name: Meirinho Holsteins LP
Address: 5418 Woodland Ave
City: Modesto **State:** CA **Zip:** 95358
Telephone: (209) 652-9780 **Fax:**
Cell/Email: akm5604@gmail.com
COPY TO: krhodes@fragservices.com
ariordan@fragservices.com
REQUESTED BY:
PROJECT:
CROP: FRESH WATER

[X] Copy of Chain [X] QA/QC Documents
Sampled By: FR Ag Services

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

(✓) DWW1: EC, NO_3-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DOM F	11/2	0835	2.7	
2	G		0840	4.1	
3	H		845	1.4	
4	I		0850	3.3	
5	J		0855	0.4	
6	K		0900	-0.6	
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Kierstan Rhodes	FR Ag Services	11/2 0900	
Second	JCF	DLV	11-2-23 2:00	11-2-23 3:45
Third	[illegible]	DLI	11/2 16:00	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[Meirinho Holsteins LP Water.xls]Dairy Water rev 3.6.18

DocuSign Envelope ID: 346F27D5-F390-4059-8426-9C642F56FE7D

11/02/23 16:00 23K0140

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	* pH Value										
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic	1	1	1	1	1	1				
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

SENT OUT